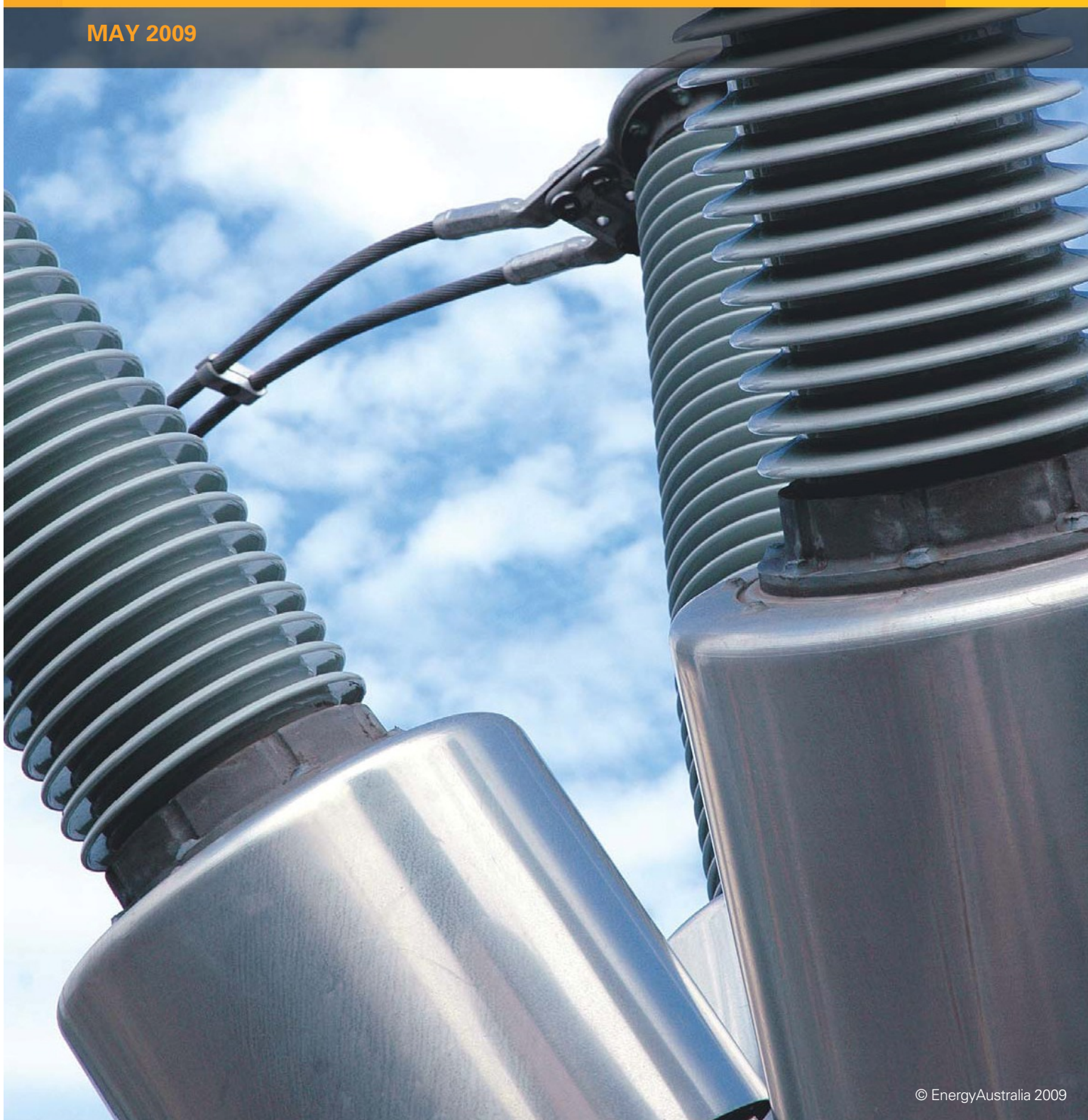


EnergyAustralia

Network Pricing Proposal (Revised)

MAY 2009



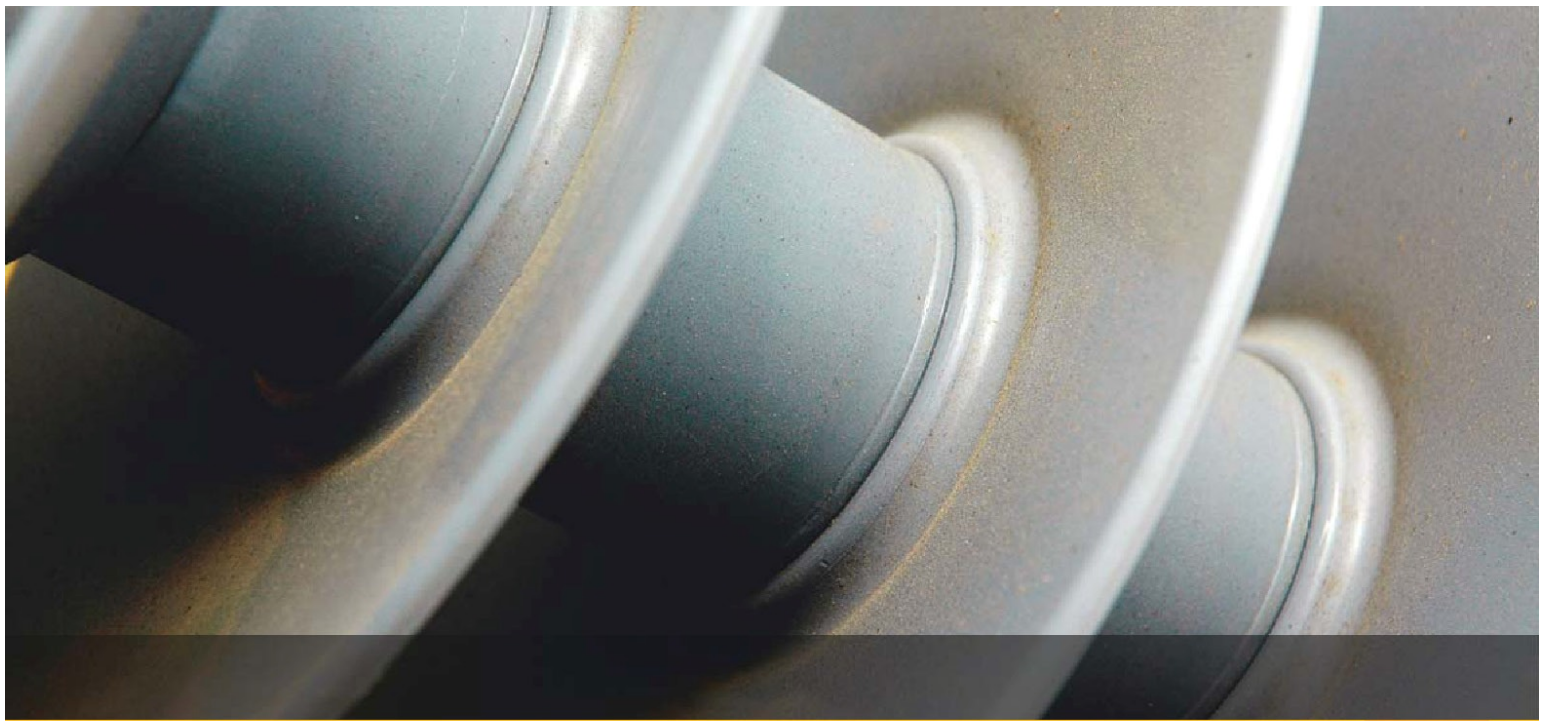


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Overview and outline

Overview and outline

Introduction

EnergyAustralia operates the largest distribution network in Australia, which services 1.58 million customers. EnergyAustralia also owns and operates a transmission support network (as a distribution business), which is in the same order of size and value as the transmission networks in South Australia and Tasmania.

On the 30 April 2009 the AER made its final determination approving the largest capital and operating investment programs in EnergyAustralia's history. These programs are required to renew the network and ensure continued energy supply to our customers. As a result of this decision, the AER approved real price increase of 17.86% for FY10 for EnergyAustralia's distribution prices. The AER also approved a real increase of 7.77% for EnergyAustralia's transmission prices.

These increases in EnergyAustralia network charges originally translated to a \$1.49 per week increase in the DUOS component of electricity costs for the typical domestic customer for FY10.¹ However, the DUoS component for a typical domestic customer in FY10 will only rise by \$0.85 per week, as a result of a revision and updating in allocation of costs both across tariffs and also between components within tariffs. When price increases for TransGrid's transmission services, and the NSW State Government's Climate Change Fund are considered, the total impact on a typical domestic customer's bill will be \$2.17 per week.

By way of background, the price faced by electricity consumers is made up of a retail energy component and a regulated network component. The energy component includes the cost of generation and retailing while the network component includes the regulated cost of electricity delivery over the transmission and distribution networks. This proposal addresses the network component only.

Document structure

This document is EnergyAustralia's Pricing Proposal for FY2010. It sets out how we proposed to set prices to recover the revenue that the AER has allowed EnergyAustralia as a result of approving our investment programs for the 2009-14 period.

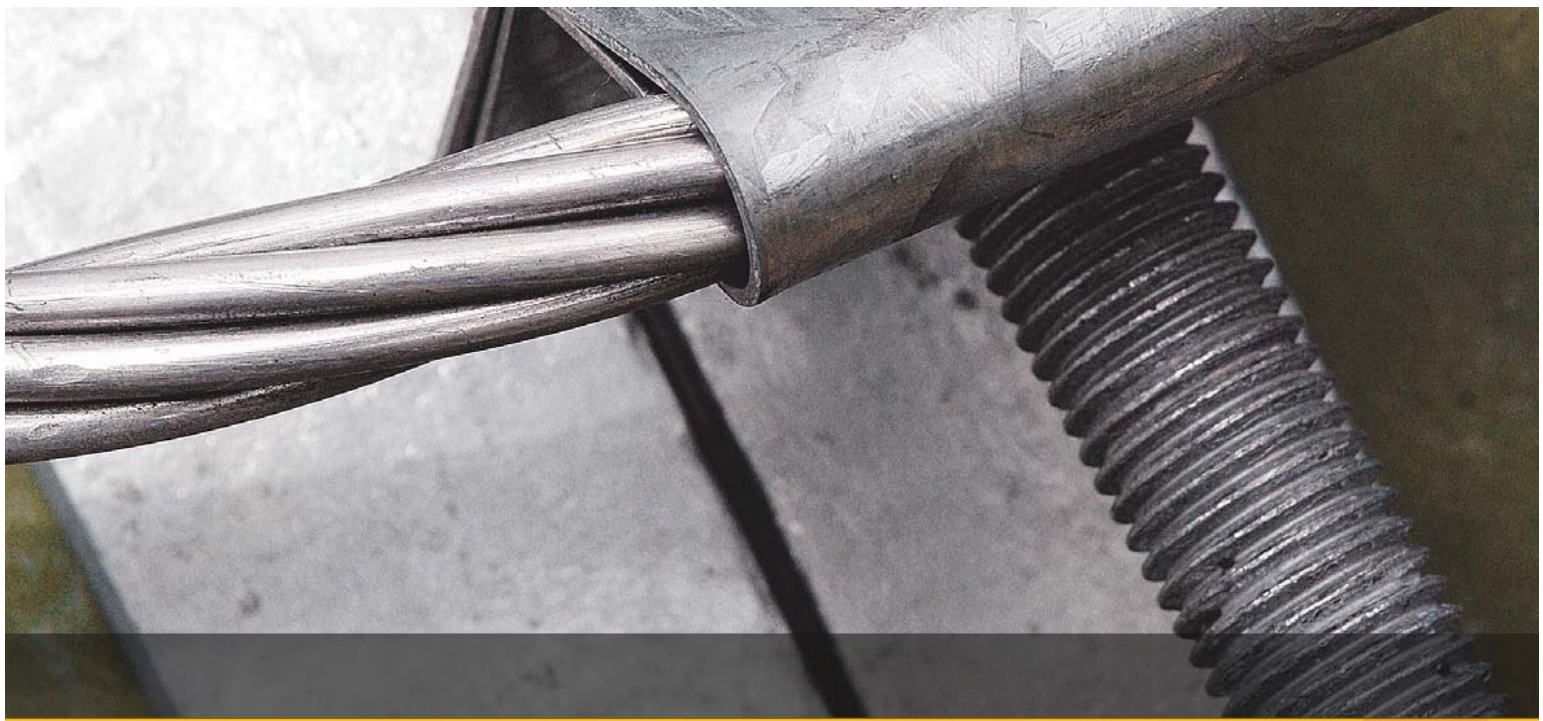
This proposal is submitted for review and approval by the Australian Energy Regulator (AER) as required by clause 6.8.2(a)(1) of Chapter 11 of the National Electricity Rules (the Rules).

The proposal is split in two parts:

- Part 1 outlines EnergyAustralia's method and strategy in setting network prices, Part 1 is written with consumers and other interested parties in mind, to assist in understanding how network tariffs are calculated.
- Part 2 specifically responds to the obligations of the Rules. Part 2 is structured to allow ready assessment of compliance by the AER.²

¹ For a median domestic customer who consumes 4700 kWh per annum, and does not have controlled load hot water. Increase includes GST and inflation. Retail related cost increases not included.

² This proposal can be downloaded from the EnergyAustralia website, www.energy.com.au/network_prices.



Part 1 – Methodology and Strategy in Setting Network Tariffs

1. Setting network prices requirement

Network price trends

Over the past twenty years network charges have fallen in real terms for all customers despite rising peak demand growth over the period. This benefit has been achieved through greater capital utilisation and an incentive based regulatory framework, which emphasises cost reduction to achieve lower prices. This reduction has also been achieved in part through technological advances in the design and construction of power system assets.

The benefits derived from increased network utilisation have largely been exhausted and parts of the network are nearing the end of their useful life. A large capital program is now required to renew the electricity network and replace the large portion of assets that were constructed during the growth phase during the 1960s and 1970s. Electricity prices are now set to rise over the next five years, to fund this substantial program of network replacement in particular. EnergyAustralia will invest \$7.88 billion³ in the 2009-14 period to meet our obligations in providing a quality network service to our customers.

The network challenge

There needs to be sufficient investment in the network to continue to be able to deliver the service outcomes expected by the community and to provide a fair return for the money invested.

Network demand peaks only last for short periods, and typically only occur on a few days each year in winter or summer. Unlike energy market peaks which occur on a global or regional basis, network peaks are highly localised.

In recent years, EnergyAustralia's network customers have chosen to install air-conditioning in over 50% of dwellings, driving higher summer peak conditions that are occurring more

frequently across the network. Despite this trend of increased demand during summer days a portion of network assets remain idle, or at best under-utilised, on all but the hottest and coldest days. This means that provision of network capacity on those peak days costs more than providing network services on other days. It is important that customers see prices that show the impact of their demand choices.

To ensure customers are provided with the necessary pricing signals to make rationale economic decisions, EnergyAustralia is investing in interval meters to move customers on to more cost-reflective Time-of-Use tariffs.

While there will remain a high degree of averaging in the pricing process for most customers, EnergyAustralia's network tariff strategy is pointing the way towards improved cost reflectivity through innovative pricing initiatives supported by advances in metering and control technology. The introduction of time-of-use tariffs, wider application of capacity charges and higher allocation of costs to peak charges are all part of EnergyAustralia's focus on improving price signals to consumers.

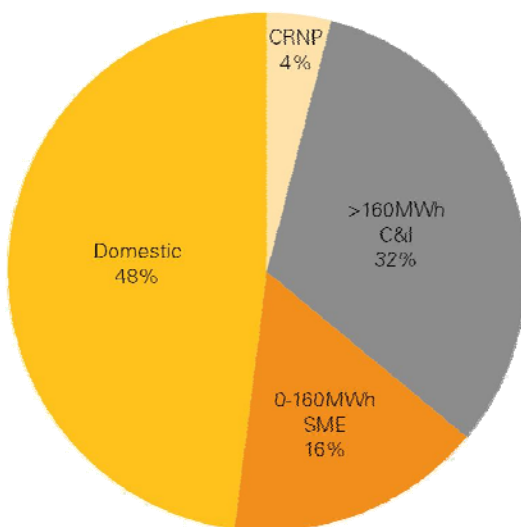
General overview

The revenue stream for the EnergyAustralia network business is derived from our customer base in the proportions shown in figure 1.

³ AER approved capital program for EnergyAustralia in real dollars at June 2009 (operational expenditure is in addition to this)

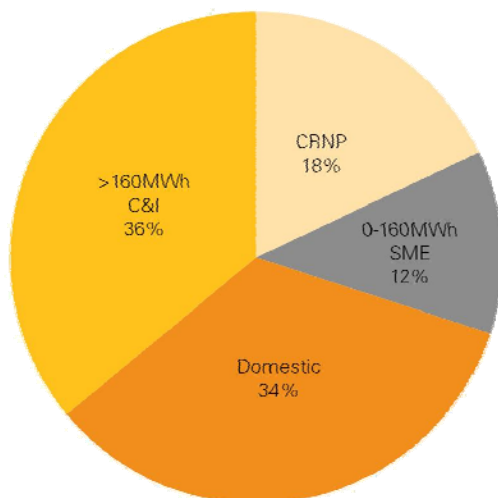
Part 1 – Methodology and Strategy in Setting Network Tariffs

Figure 1 FY08 Revenue by Customer Segment



As can be seen from figure 1 and figure 2, EnergyAustralia's largest cost reflective customers represent 18% of energy volumes but only represent 4% of revenues. Conversely, domestic

Figure 2 FY08 Energy by Customer Segment



customers only consume about a third of the total energy volumes, but account for almost 50% of revenue. This is generally reflective of the amount of assets required to service these customers. For instance, large customers, though they use large volumes, are connected high up in the network and attract smaller average customers use the electricity network all the way down to their LV connection. Their average network price is therefore high, and the total revenue retrieved from low voltage customers is higher as a result.

Table below outlines EnergyAustralia's network tariff strategy across the regulatory periods from 1999 through to 2014

As can be seen, EnergyAustralia has been progressing towards simplifying and refining pricing structures to improve their effectiveness. A substantial amount of reform has taken place during the 2004-2009 regulatory period:

1. Introduction of ToU pricing on a wide scale to small business and domestic customers
2. Increases in peak energy and peak capacity charges to improve the strength of peak period signals
3. Simplification of capacity charge to be on a rolling twelve month window, rather than on a fixed 24 month window with a complex reset arrangement
4. Rationalising the use of demand and capacity charges down from six components to one peak capacity charge component. This has greatly improved customers understanding of peak costs by making the tariff more understandable. It has also increased the financial incentive for reducing peak usage.
5. The introduction of capacity charges for business customers who consume as little as 40 MWh per year. This is equivalent to a maximum demand of 15 kVA (most utilities only apply demand charges down to 100 kVA to 250 kVA customers)

The intention has been to provide a clear and simple signal that reflects peak costs, spearheaded by EnergyAustralia's ToU program. Table 1 shows the ToU program is discussed in more detail.

1. Setting network prices requirement (continued)

Table 1 Summary of Tariff Strategy

Control Mechanism	Segment			
	Residential (0-40MWh pa)	SME Business (0-160 MWh pa)	C & I Business (Above 160 MWh pa)	CRNP (10MW + Demand)
1999-2004 Revenue Cap	Single energy rate with fixed daily charge	Single energy rate with fixed daily charge	Highly sophisticated 10 part tariffs with multiple demand and capacity components	Legacy tariff structures
2004-2009 Price Cap	↓ Two part energy rate with usage above 7MWh p.a. higher than 1st block, with fixed daily charge	↓ Two part energy rate with usage above 10MWh p.a. higher than 1st block, with fixed daily charge	Simplified structure with one peak capacity component supporting Peak/Shoulder/Off Peak energy components	Legacy tariff structures
	Migrate Time of Use tariffs with interval meters	Migrate Time of Use tariffs with interval meters		
2009-2014 Price Cap	Consider the impact of current structure, on customer peak use before proceeding with further reform	Consider the impact of current structure on customer peak use before proceeding with further reform	Retain structure from 2004-09 period, review effectiveness of tariff reform in managing peak use	Align CRNP tariff structures to standard customer class tariffs

2. Pricing initiatives

Over the last 15 years, EnergyAustralia has been a national leader in the area of promoting more cost reflective tariffs. EnergyAustralia's pricing initiatives have been directed at influencing customers to reduce their peak period energy consumption. The intended outcome is to defer capital expenditure on both network and generation capacity. This in turn reduces costs to all customers. The inherent incentive to a Distribution Network Service Provider is that with capital expenditure locked in under an ex-ante framework, capital deferral translates to savings to the business during the regulatory period. Two key initiatives, our ToU program, and our Strategic Pricing Study reflect our desire to find ways to reduce peak costs.

2.1 EnergyAustralia's ToU electricity pricing program

EnergyAustralia is at the forefront of utilities worldwide, in having mandated Time of Use (ToU) pricing for all new and upgraded supply connections and in its roll out of interval meters and ToU pricing. ToU electricity pricing has been in place for many years for larger business and industrial customers. However, the cost of electronic interval meters has decreased in recent years, and EnergyAustralia no longer purchases rotating disc meters. Since 2004, all meters purchased for new connections and replacement purposes have been interval meters, permitting ToU pricing to be extended to smaller customers.

EnergyAustralia's ToU Program contains the following main elements;

- Disc meters at the majority of medium sized customer premises (in the consumption range of 15-160 MWh per annum) have been replaced by interval meters. At the writing of this proposal (May 2009), this meter replacement program was approaching a June 2009 completion;
- Customers with interval meters have been transferred to ToU prices
- All new and upgraded customer connections receive an interval meter and are placed on ToU pricing. This includes new domestic dwellings as well as businesses.

The Independent Pricing and Regulatory Tribunal, in its 2004 Determinations, facilitated this ToU rollout by allowing expenditure for meter replacement and modifying the pricing side constraints to allow ToU prices to be mandated.

As at May 2009, EnergyAustralia has over 227,000 network customers on ToU tariffs (on a customer base of close to 1.6 million), with 176,000 of those facing EnergyAustralia Retail ToU rates⁴. By June 2014, it is anticipated that in the absence of Advanced Metering Infrastructure, over 500,000 customers will be on network ToU tariffs, making it easily the largest deployment of interval meters in Australia.

Effectiveness of the Time of Use Electricity Pricing Program

The effect of ToU pricing on customer's consumption patterns is expected to build up over the longer term. As at May 2009, only a mild response of a 1.1% demand reduction co-incident with the network system peak has been measured. EnergyAustralia plans to regularly review and update the performance of ToU over the coming regulatory period.

The key messages of the ToU program have been:

- A higher rate is charged during "Peak" hours when electricity demand is high and lower rates are charged during the "Shoulder" and "Off-peak" times when the demand is lower;
- Customers can save money on ToU;
- Customers pay for what they use - ToU effectively stops lower energy users subsidising customers who use large amounts of energy at peak times - they have previously been charged the same rates;
- ToU metering is designed to spread electricity usage over the whole day and therefore reduce peak demand for electricity. This is known as 'demand

⁴ The remaining 50,000 customers are with an external retailer, and EnergyAustralia is aware that roughly only half these customers see retail ToU rates.

2. Pricing initiatives (continued)

management', and is a cost-effective alternative to just building more and more infrastructure;

- Customer can save money by making simple changes to their use of electricity such as running their dishwasher out of peak hours and installing time switches where appropriate.

Recent analysis by EnergyAustralia of 32,000 household electricity accounts already on Time-of-Use billing found families were saving on average \$64 a year compared to standard flat pricing. We also found that on average families were using 78% of power outside peak times. Additionally, 69% of customers were better off under time-based pricing and 71% of customers surveyed said it was a fairer pricing system.

Smart meters are a new way of measuring electricity use that give customers more control over their electricity costs. They can also help customers save money on their electricity bills by encouraging them to change some of their non-essential electricity use to outside peak times.

In addition to the mandated transfer of customers to ToU, since July 2008 an additional 2,640 customers have voluntarily asked EnergyAustralia to install a smart meter to help them manage their electricity bill.

2.2 EnergyAustralia's strategic pricing study

Our Strategic Pricing Study (SPS) has been designed to quantify how customers would respond to tariff reform initiatives such as Seasonal and Dynamic Peak Pricing.

The principal objectives of our SPS were to:

- Test customer price elasticities for specific product offerings aimed at defined target customer segments;
- Assess the impact of specifically prepared information materials on customer perception and behaviour;
- Test product market and take-up potential; and

- Experience in the implementation and management of customer communications.

EnergyAustralia's SPS largely adopted the California State-wide Pricing Pilot (SPP) approach, modified for local conditions and EnergyAustralia specific business objectives.

The Dynamic Peak Price (DPP) tariff is a highly cost reflective tariff, where retail and network prices increase significantly (10 to 20 times) for the top few hours of peak demand each year. Outside these hours, DPP tariff prices are cheaper than default tariffs, to target revenue neutrality on an annual basis. Customers save money if they reduce load in DPP periods.

The peak periods can be 'called' at short notice, and the customer is informed from 2 to 24 hours in advance that a peak event will take place. Customers in the trial were informed of peak events via SMS, email and auto-phone messaging.

The DPP trial measured tariff take-up by customers, and the peak load reductions from dynamic price signals. The study also compared the customer response between business and domestic customers, different price points and information only, and between customers with and without in-house displays.⁵

Customers were randomly sampled to participate in the study, which ran for 2.5 years. The customers were grouped into five groups and a control group:

1. Seasonal Time of Use
2. Dynamic Peak Pricing Medium with In-house Display
3. Dynamic Peak Pricing High
4. Dynamic Peak Pricing High with In-house Display
5. Information Only
6. Control

⁵ In House Displays (IHD's) display information on electricity consumption and prices to the consumer

Part 1 – Methodology and Strategy in Setting Network Tariffs

This allowed EnergyAustralia to test the following propositions:

- What is the optimal price point for dynamic prices?
- What is the additional benefit of an In Home Display?
- What is the relative demand response of domestic versus business customers?
- Is the demand response predominantly a shift or simply a reduction in peak periods?
- Was the response more driven by information than price?
- What role does weather play?
- What is the cost versus benefit of deploying such tariffs?
- Will customers take up such tariffs, and respond to them at meaningful levels?

EnergyAustralia is in the final stages of analysing the results from the study and detailed results and conclusions are planned to be made public around September 2009.

In summary, EnergyAustralia found that domestic customers on the trial responded by up to 36% to peak events on hot summer days. Business customers unfortunately showed no meaningful demand response. The results are being refined to determine whether the voluntary take up of such tariffs can be achieved in sufficient numbers and concentration at a particular location to make this tariff a useable demand management tool.

A summary of the results to date are shown in Table 2.

Table 2 Residential Response Levels to Dynamic Tariffs

Tariff	Average across all DPP events	Extreme temperatures		Assessment
		Winter (5°C)	Summer (31°C)	
Information only	11%	11%	13%	Inconsistent, ranges from -1% to 24%
DPP Medium with IHD	25%	30%	36%	No difference in response between medium \$1/kWh and high \$2/kWh prices
DPP High with IHD	25%			
DPP High without IHD	23%			IHDs add minimal response

The key learnings from the study have been:

- Domestic customers who sign up to the tariff respond well to dynamic tariffs, and are willing to reduce their air conditioning usage on hot days, despite popular belief to the contrary. The further away from the comfortable 18° to 22° Celsius range, the greater the demand response.
- Despite the apparent complexity of dynamic tariffs, customers are able to understand the concept
- The dynamic price could probably be set at 50-70c/kWh and still achieve comparable reductions. The demand response to prices above \$1/kWh appears to be saturated.
- 0-160MWh p.a. business customers show no response to dynamic tariffs. There may be business customers above 160MWh p.a. who would respond to such tariffs

These results demonstrate a potential for tariffs to play a role in demand management. There are a number of questions that require resolution before proceeding to an operational deployment of such tariffs. In particular is the customer take up of such tariffs on a commercial and voluntary basis. This drives the potential demand response at a particular location since it is the aggregate reduction that is important. This is difficult to quantify from the Strategic Pricing Study, because

2. Pricing initiatives (continued)

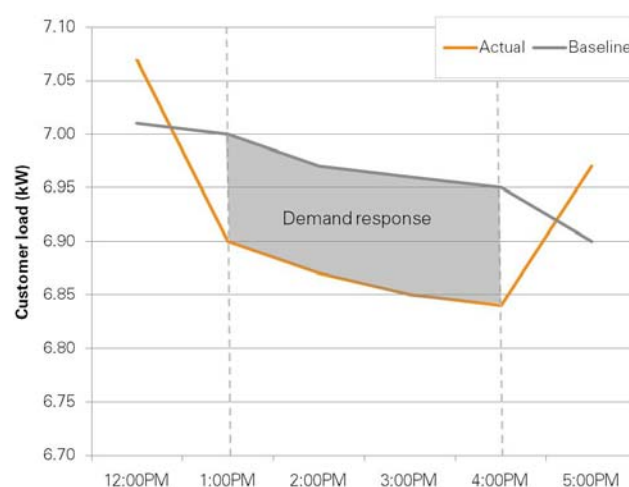
being a trial, it is unclear exactly how the take up rates from the study reflect what would happen in a commercial deployment.⁶

2.3 Proposed new tariff trial during 2009-14

EnergyAustralia is planning to build on the results of the Strategic Pricing Study by launching a new tariff trial during 2009-14. The trial will examine rebate tariffs as a method of achieving demand response. By offering a carrot rather than a stick, it is anticipated that there will be a much greater take of this type of tariff, improving the expected aggregated demand response. The rebate arrangement is also operationally much simpler to implement, vastly improving its potential as a real tool for deployment within the next five years.

The rebate tariff works by paying customers for load reductions. The more they reduce their usage, the more they are paid.

Figure 3 Example of Rebate based off a Load Baseline



2.4 Assessing the performance of network tariffs

EnergyAustralia has carried out substantial reform on our network tariffs during 2004-09. To this end, it is now appropriate to assess and review the performance of these reforms. Do these prices have a meaningful impact on customers' behaviour? Do they make sense? Do we need to consider better education and marketing? What role do retailers have in promoting our tariffs?

- EnergyAustralia intends to embark on an ongoing program of review, that will attempt to provide effective feedback on the performance of our tariffs. This will be driven by two key initiatives:
- A quantitative sampling program assessing customers demand response to ToU tariffs. This program will provide key data on levels of peak reduction and/or shifting over time across the full range of customer sizes and types
- A qualitative program seeking customer feedback on their understanding of their tariff structures and their general levels of satisfaction

⁶ A key consideration is the Hawthorn effect where customers, knowing that they are going on a trial, improve their performance. The ideal trial would be one that replicated the exact conditions of a commercial deployment around a constrained zone substation.

Part 1 – Methodology and Strategy in Setting Network Tariffs

Taken together, these initiatives will help determine who is responding to ToU tariffs, and how they are achieving reductions. This will then allow refinement of ToU structures and also assist in determining which customers to target with education material and potentially more sophisticated voluntary dynamic ToU tariffs.

Importantly from a price setting process, this feedback will provide key insight on whether prices are set at optimal levels and structured in the most efficient manner.

2.5 Smart metering infrastructure

EnergyAustralia has also been at the forefront in assessing the value of smart meters and was the first utility in Australia to trial communications enabled smart meters with a 7,000 meter deployment. The trial tested a variety of communication technologies from a number of vendors, and also provided valuable insight into the cost of deploying such technology.

We are also developing a showcase of smart grid technology in the suburb of Newington, near the 2000 Olympics site at Homebush Bay. The project will include a home area network that can communicate the usage details of air conditioners (and other appliances) to customers. The customers will have full access to their consumption information via a web portal. We will also use the showcase to carry out initial trials of our proposed rebate tariff before proceeding to a full trial.

3. Price setting process

In setting prices, EnergyAustralia has to consider the constraints of the weighted average price cap and also the proposed prices from our cost allocation method (generally known as 'cost reflective' prices). It is important to understand that the price cap and the prices from our cost allocation methodology don't immediately reconcile.

The simplest approach to setting prices would be to ignore the cost reflective prices and simply escalate all current components by the allowed X-Factor. Conversely EnergyAustralia could simply attempt to move customers directly to cost reflectivity and ignore the constraints of the weighted average price cap. Of course both approaches would be problematic, with customers prices either drifting away from cost reflective prices over time (simple escalation) or receiving severe price shocks (cost reflective). In addition, it is quite likely that immediate application of cost reflective prices may well breach the price cap and side constraint obligations under the National Electricity Rules.

The approach adopted by EnergyAustralia is to take account the cost reflective prices, and move toward them over a number of years. This approach balances the need to charge users for their fair use of the network, but also allows time for customers to understand and respond to those charges. Prices are then adjusted to ensure they meet the constraints of the price cap. This may require slowing down the transition to cost reflectivity.

The following discussion outlines how EnergyAustralia sets prices, by first considering the constraints of the weighted average price cap, and then outlining our cost allocation methodology. Finally there is a brief discussion as to how EnergyAustralia has reconciled the price cap with the reflective prices recommended in our cost allocation model.

3.1 Setting the constraints

The AER's 2009-14 Determination specifies a Weighted Average Price Cap (WAPC) control mechanism for standard distribution control services. The WAPC limits price movements over the regulatory period, so that the business receives revenue gains or losses depending if volumes rise or fall. The WAPC works by weighting each proposed price by its historic volumes, and dividing this result by existing prices weighted by the same historic volumes. This "weighted

average" price increase is limited to a combination of CPI, X factor and D factor for that year:

$$\frac{\sum_{i=1}^m \sum_{j=1}^n p_t^{ij} \times q_{t-2}^{ij}}{\sum_{i=1}^m \sum_{j=1}^n p_{t-1}^{ij} \times q_{t-2}^{ij}} \leq (1 + \Delta \text{CPI})(1 - X_t)(1 + D_t)$$

where:

- p_t = the proposed prices for the upcoming financial year
- p_{t-1} = prices for the current financial year
- q_{t-2} = volume quantities from the most recently completed financial year
- X_t = the allowed real average price increase set by the AER from their April 2009 Determination
- D_t = the AER approved adjustment to recover costs from successful demand management initiatives
- ΔCPI = the most recent historic average December to December change in CPI

For 2009, the change in CPI is 4.35%, the allowed X-factor is 17.86% and the D-factor is 0.00%. Distribution prices are therefore allowed to increase on average by 22.99% for 2009/10.

3.2 Basic steps in price setting

There are four basic stages in developing Distribution Use of System (DUoS) charges:

1. Establish medium term pricing policies that will help achieve EnergyAustralia's overall business goals. Identify appropriate paths from existing price structures to the desired ones.
2. Establish the weighted average price movement that network tariffs must not exceed.
3. Allocate the permitted annual price change to customers based on their use and impact on the network.

Part 1 – Methodology and Strategy in Setting Network Tariffs

4. Develop pricing structures that optimise revenue in a manner that is equitable, minimises risk, and, as far as possible, reflects the marginal costs of network usage.

It is essential to take at least a medium term view of revenue recovery and pricing. In this way, customers can be given a smooth path from one pricing regime to another and time to understand why prices are changing and to plan ahead to manage their electricity costs.

Distribution businesses must also recover the cost incurred for use of the transmission network. These costs are recovered as transmission charges on top of distribution charges. To recover these costs, the AER has allowed a pass through mechanism

of transmission costs. Where the distribution business under or over recovers transmission revenues compared to the transmission costs it incurs, the difference is carried forward as an adjustment in the following years prices

- This remainder of this chapter outlines EnergyAustralia's goals for price setting and the principles it applies in achieving these goals. These goals may at times diverge, requiring strategies that balance the outcomes to consumers and stakeholders. These goals and principles are summarised in Table 3.

Table 3 Pricing goals

Goals		Description
Achieve economic efficiency through: Dynamic efficiency Allocative efficiency		Use resources in an optimal way, there are two sub-goals: Pricing to encourage efficient supply and customer side investment & optimise productivity Allocate full costs accurately to the customers who incur them.
Achieve regulatory entitlements		The regulator determines the maximum allowable price movement to cover smoothed capital, operating and depreciation costs.
Encourage economic behaviour by customers		Prices that reflect costs provide signals that enable market forces to operate most effectively. It allows each customer to decide if a price signal is sufficient for them to modify their electricity consumption.
Satisfy customers	Comprehensible	Customers should be able to identify how their network charges are calculated and identify the implications of their usage.
	Equitable	Prices must treat equally those customers who have the same usage profile; in addition prices should be the same for all retailers
	Stable	Customers may not be able to manage sudden upward changes in price so when a price needs to increase, the change is usually phased in.
Manage demand		Networks sometimes seek to constrain demand for a variety of cost and environmental reasons and pricing can assist this action.
Improve cost allocation		Identify as closely as possible what it costs to provide a network service to each customer or group of customers supplied by the network
Refine price structures		Improve economic signals to match the various cost components of the network.

3. Price setting process (continued)

3.3 Achieve economic efficiency

Network businesses are highly asset intensive and the asset related costs form the major proportion of total costs. It follows that the main determinant of efficiency is how effectively EnergyAustralia Network invests in the network to provide services to customers.

EnergyAustralia Network seeks to be an efficient user of resources in all areas of its operation. It is essential to continue to minimise costs in day to day operations and also ensure optimal benefit from the investment in network assets. To deliver this, EnergyAustralia Network will continue to improve its analysis and the modelling used to determine the costs and benefits that apply to different customer groups. This provides the basis to set cost reflective prices for different customers or customer groups. In turn, customers can then make their own rational investment decisions. Network pricing takes into account two types of efficiency.

Dynamic efficiency

EnergyAustralia Network achieves dynamic efficiency through continuous productivity improvements, innovations and efficient use of resources. EnergyAustralia Network continually evaluates new technology and ways of operating the business that may reduce costs. The AER restricts price increases (and the total permissible revenue) so as to protect customers and to provide DNSPs with an incentive to increase productivity (since prices are capped, networks can only improve profit margins directly by being more efficient).

Allocative efficiency

In efficiently allocating costs, it is important to distinguish between the fixed and marginal cost components of operating and maintaining an electricity network. Fixed costs are independent of load while marginal costs are associated with carrying the next increment of load. We must then distinguish between short and long term marginal costs.

A discussion on Short Run Marginal Cost (SRMC) assumes no new infrastructure investment is required. A DNSPs short run marginal costs are usually very small – the poles and wires are in place and it costs little extra to carry a marginal increase in energy. Increased costs are usually limited to electrical losses

and the cost of more expensive generation, where the capacity of the network does not permit the lowest cost generation to be used (termed out-of-merit generation). Both of these short run costs are traded within the electricity market and form part of the energy cost, rather than the network price.

Long Run Marginal Cost (LRMC) takes into account future investment for network growth. Network assets have very long lives, generally in excess of 30 years, and often, long construction times. Network augmentation is thus extremely 'lumpy' as there are large, high cost investments at irregular intervals. Pricing of network services on a LRMC basis is appropriate to this situation. Regulated monopolies have traditionally considered the life of an asset in financing it and signal to customers the cost of future development of network infrastructure. A full discussion on LRMC is contained in Part II of this Price Proposal.

Only a long-term view of the network planning requirements and associated cost recovery will lead to sustainable development and funding of the network. This marginal cost approach is important in achieving allocative efficiency but as there will generally also be a need to recover fixed costs (which do not contribute to the marginal cost); the marginal cost approach alone will not recover sufficient revenue.

3.4 Achieve regulatory entitlements

To ensure the long-term viability of the network, regulators permit EnergyAustralia Network to establish weighted average price movements that are expected to generate sufficient revenue to cover changes in depreciation, capital and operating costs and over the regulatory control period. This type of regulatory framework is also known as a weighted average price cap which is discussed in Section 1.6.1.

Pricing structures are developed to optimise revenue recovery within the confines of the WAPC. For example, pricing strategies could increase fixed charge components to help mitigate volume risk under the WAPC (subject to the price limit and fixed price side-constraint). Under the previous revenue cap, there was no volume risk as over or under recovery of revenue was accounted for in subsequent years. Also, the AER has set out requirements for managing the 'under and over' recovery of transmission revenue in its Determination.

Part 1 – Methodology and Strategy in Setting Network Tariffs

Price changes are required to meet changing business costs in a number of areas. The four main components of regulated

business costs are shown in Table 4.

Table 4 Network revenue components

Service	Description
Transmission	TransGrid calculate charges for the whole transmission network in NSW. The transmission network includes their own assets, together with transmission assets owned by EnergyAustralia, Country Energy, and a 110kV DC NSW/QLD interconnector owned by APA Group. The transmission charges are paid by distributors as well as a small number of very large customers. Distributors then recover the costs of the transmission network by adding on these costs in distribution charges faced by end users. The transmission network connects large generators together, and transports the energy they produce to distributors.
Distribution	Customers pay a use of system charge to meet the cost of using the distribution network.
Climate Change Fund	EnergyAustralia is required to make payments to the NSW Government Climate Change Fund. This provides funding to promote energy efficiency and renewable energy technology. The cost of the fund is recovered through a small uplift on the network tariffs faced by end users.
Ancillary activities, eg monopoly charges, miscellaneous fees.	There is a range of miscellaneous activities for which EnergyAustralia charges fees, such as extra meter readings, new connections, answering conveyancing enquiries. These fees are regulated and form part of the WAPC.

3.5 Encourage economic behaviour by customers

EnergyAustralia wants customers to value the service they receive. If a service were to be provided at less than its economic cost, it would encourage wasteful use.

Pricing is used to encourage customers to constrain their consumption or take up services that are less costly to deliver. By sending price signals, customers will be able to make choices about their level of electricity use. For example, off peak hot water is much cheaper than normal prices. This signals that off-peak controlled load places minimal stress on the network and would encourage a shift in consumption to this period. Charging higher prices for access to the network at peak times encourages customers to limit their consumption at these times. If they do not, they must accept higher prices, which would in turn provide the revenue to upgrade the network to be capable of handling the higher peak loads.

3.6 Satisfy customers

EnergyAustralia places considerable emphasis on consumer input when determining network prices. Proposed changes in prices are discussed with the Australian Energy Regulator, retailers, consumer advocates and major customers so that the customer perspective is properly considered.

EnergyAustralia hosts an annual Information Forum for retailers to advise them of developments in network pricing and other matters of interest to them and the customers they represent.

It is inevitable in a period of change that some customers may be concerned that they are being disadvantaged. A key principle is that there should be equity in pricing. Important in this is the concept of user pays. There is never any intent to disadvantage any group of customers.

As well as the overall WAPC, the Rules place side constraints on the movement of distribution component of prices faced by consumers. Prices cannot move more than 2% above the average price increase for that year for each tariff class. That is, if the distribution component of prices is allowed to rise by 10% in a given year, each class of tariffs cannot increase by more than 12%. It should be noted that the transmission

3. Price setting process (continued)

component of network prices is not subject to this constraint, so consumers can still see a price increase beyond this 2% rule if transmission prices increase dramatically from one year to another. However, transmission prices only make up about 15% of network charges, so they do not have much an impact on overall network prices for small customers.

Make prices comprehensible

Ideally, all customers should be able to understand how prices are determined and how their bills are calculated. Unfortunately, simplicity is not always consistent with achieving other goals. Developing prices which are as equitable as possible, which encourage economic behaviour and which ensure the right return on investment would lead to quite complex price structures. How the actual figures for each price component are determined can be similarly complicated. A major aim of this report is to explain the issues involved.

More complex prices have tended to apply to larger customers such as businesses, while domestic customers – who comprise almost 90% of the total customer base – have had fairly simple price structures. However, the highly averaged pricing signals which domestic customers have historically been charged is unsustainable. Over the next few years, EnergyAustralia will be introducing more sophisticated options for domestic customers, in order to deliver cost reflective pricing signals. The challenge will be to develop prices that are comprehensible, and EnergyAustralia Network is committed to ensuring that everybody understands their options.

Maintain stable prices

Customers do not want to be subject to highly fluctuating prices and should be able to anticipate the cost of their electricity supply. In addition to the general X-factor constraint on the movement of prices the Rules constrain individual prices by side constraints of 2% beyond the X-factor increases.

It is important for stakeholders to understand that while whole of price movements will be limited on average by side constraints as set down in the Rules, the impact on any given customer may be higher or lower than the average. EnergyAustralia is substantially increasing its customer research activities to better anticipate pricing impacts of proposed new tariffs or rebalanced tariffs (among pricing components). Any tariff change that is expected to result in a

material price movement for customers will be accompanied by a transitional arrangement to smooth the impact over time.

Make prices equitable for all customers

EnergyAustralia seeks to ensure that customers are treated fairly. While two similar customers next door to each other would normally be charged on the same basis, customers whose location or consumption pattern costs the DNSP more to service should expect to pay a greater cost.

This is in part achieved through capital contributions, which supplement prices for the use of the network. The capital contributions policy was determined by IPART in 2002⁷. Each new customer is required to pay for the installation of dedicated connection assets and in some cases assets that would be shared by other customers. Thus, the cost of a long connection to premises distant from the street mains would be met up-front.

3.7 Manage demand

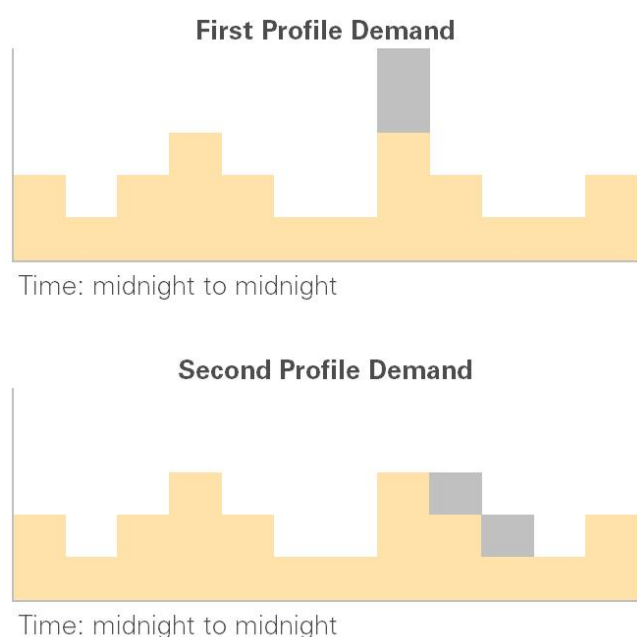
Demand Management (DM) focuses on reducing peak demand to improve network utilisation, and mitigate the need for augmentation. It is important to note that network DM is different in nature to other forms of electricity demand management, reflected in the pricing of network DM benefits, the value of which is localised and transient. This is significantly different to the aim of energy based demand management, which focuses on reducing resource consumption and the associated greenhouse gas emissions through enhanced efficiency measures.

The simple load profiles in Figure 4 shows how demand may be moved from the peak to times when demand is lower. In this simple illustration, the network need have only 60% the capacity to supply the second profile. Pricing strategies target this demand management goal.

⁷ Capital Contributions and Repayments for Connections to Electricity Distribution Networks in New South Wales - Determination No 1 2002, IPART, April 2002

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Figure 4 Reducing capacity by transferring demand



Equally, a customer that consumes energy mainly at off peak times should receive a lower network charge than one that consumes at peak times, because it is using the network within readily available capacity and not creating the need for extra investment. Thus, the basic principle that recurs throughout this report is one of user pays. This is the basis for ensuring equity between customers, although in some cases preferred prices may need to be moderated to meet customer expectations.

3.8 Improve cost allocation

The 'user pays' principle can encourage rational economic behaviour and be equitable. In recent years, EnergyAustralia has been refining its 'Network Tariff Model' for allocating costs to customer classes. Such refinement will continue so that each customer pays an amount which is as close as possible to the costs that EnergyAustralia incurs on the customers' behalf.

3.9 Refine price structures

No single pricing component can deliver an optimal pricing signal, as the cost allocation process generally results in prices that would exceed the marginal cost of supply. Thus a minimum of a two-part price structure, containing both usage-related and fixed components, is required to signal to the customer the impact of their use of the network. By combining additional pricing components in a more sophisticated structure, it is possible to get much closer to achieving desired pricing outcomes.

4. Cost Allocation

The first step in setting tariffs is to determine the cost of supplying various customer classes. In theory, it might be possible to determine prices for each individual customer, but it is neither cost effective nor practicable to do so, except for the largest customers. Instead, most customers are grouped into classes according to their level of connection to the network and consumption pattern. If a customer is the only one using a particular asset, it should meet the associated costs, likewise if a customer class is the only class using a group of assets, the customers in that class should share its costs. If multiple classes utilise an asset type then all the classes' members will share (but not necessarily equally) the costs. The multiple asset cost pools are shared in various proportions by multiple customer classes.

Industrial companies have very different consumption patterns from suburban homes. The classes are based on several factors:

- The voltage level of the supply;
- The point of connection (ie. within the network or at the upstream substation);
- The nature of the usage.

Householders are only supplied at low voltage while businesses are supplied according to their needs and the available mains supply. Some businesses are supplied at high voltage (11 kV) and very large ones at subtransmission level (33 kV to 132 kV). The costs associated with each type of supply are different and allocated accordingly.

The nature of the usage is classed as business, domestic, or unmetered. The different usage patterns have markedly different effects on the operation of the network.

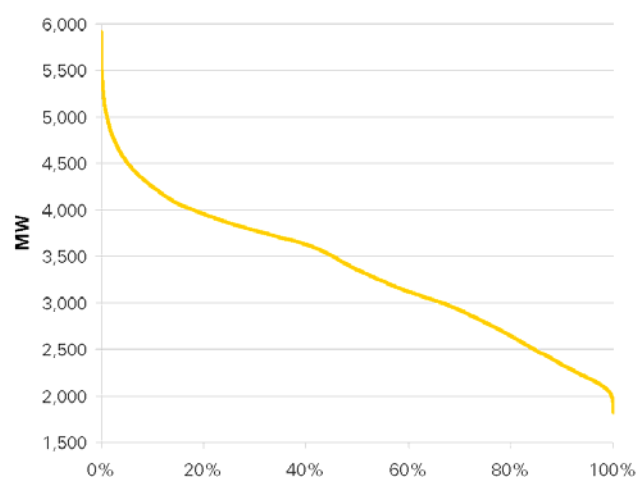
4.1 The cost of network expansion

The concept of the network load duration curve can be used to provide a broad indication of the risk that the capacity of the network may fall short of supplying the required load. However it is important to note that network planning decisions are made on location specific load data, not total system aggregates.

The load duration curve in Figure 5 illustrates the proportion of time for which the system load exceeds a particular threshold.

Such an averaged approach can be used to develop averaged prices for classes of customers.

Figure 5 EnergyAustralia system load duration curve (for the 12 months to 30th April 2009)



Top 10% of network load (5326 MW) is utilised less than 0.27% of the time
Second 10% of network load (4734 MW) is utilised less than 2.67% of the time
Bottom 30% of network load (1775.4 MW) is used 100% of the time

This load duration curve shows for example that the total system load is greater than 5,000 MW for 1.076% of the time, or (1.076% x 8760 = 94 hours per year). If the total load on the network were to be limited to 5,000 MW, load curtailment might be required for that number of hours per year, when the total load was in excess of that value. (A location specific analysis would need to be carried out for any results that could be used for demand management or network planning.)

Load duration curves reinforce the fact that throughout the network much of the existing network capacity is underutilised. Network pricing should provide incentives for customers to utilise network capacity as efficiently as possible.

Allocating fixed costs

The electricity network is sized to meet the maximum demand of consumers, so it would seem appropriate to allocate the sunk costs of the network based on each customers' contribution to this peak. There have been two basic approaches to this allocation problem:

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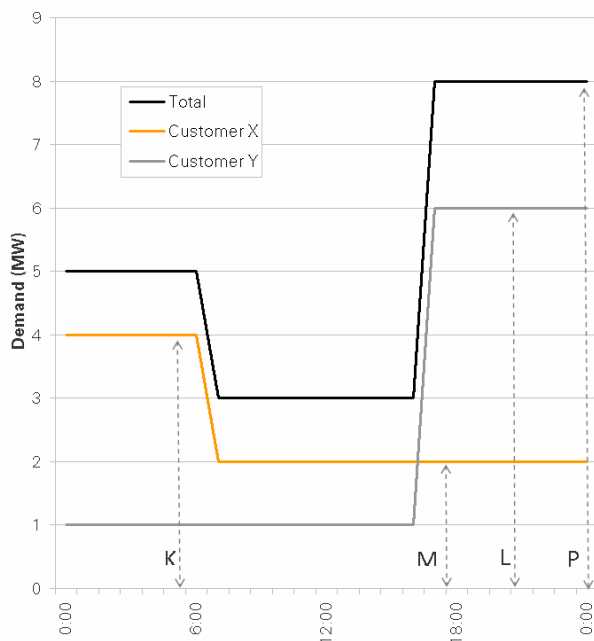
Method A: Allocate in proportion to a consumers maximum demand, regardless of the time of its occurrence

Method B: Allocate in proportion to a consumer's maximum demand co-incident with the network peak, regardless of the consumers maximum demand (which may be at another time)

Consider two customers X and Y with loads as shown in figure 5. Customer X has a maximum demand of K kW, while customer Y has a maximum demand of L kW. The combined demand is P kW.

At the time of the maximum demand, customer Y's maximum demand is still L kW, but customer X's demand at this time is now M kW

Figure 6 Load Profiles of Two Consumers



Using the two allocation methods outlined above:

Table 5 Example allocation

	Method A	Method B
Allocation to Customer X:	$= K/(K + L)$	M/P
Allocation to Customer Y:	$= L/(K + L)$	L/P
Total	$= 1$	$(M + L)/P$

Customer X would benefit from Method B, while Customer Y would benefit from Method A.

Method B can be defended on the basis that since peak costs drive costs, then it is the contribution to the peak that matters. Any consumption outside this peak period is irrelevant and does not drive cost.⁸ One of the key problems with this method though is that as demand profiles change over time, and for example customer X's load grows significantly in the mornings, then the peak demand can immediately shift. Costs then suddenly need to be reallocated. Method A does not have this problem.

The difficulty with Method A on the other hand is that it has limited strength in signalling the cost of supplying the network. While it uses peak demand, these may not relate to the cost drivers of the network. Why should someone be penalised for off peak use if it doesn't contribute to peak?

A third way is therefore proposed. This approach is known as the method of intercepts, and continues to allocate based on a balance of each customers contribution to the peak, but balanced against their use outside of peak times as well.

The basic method of intercepts

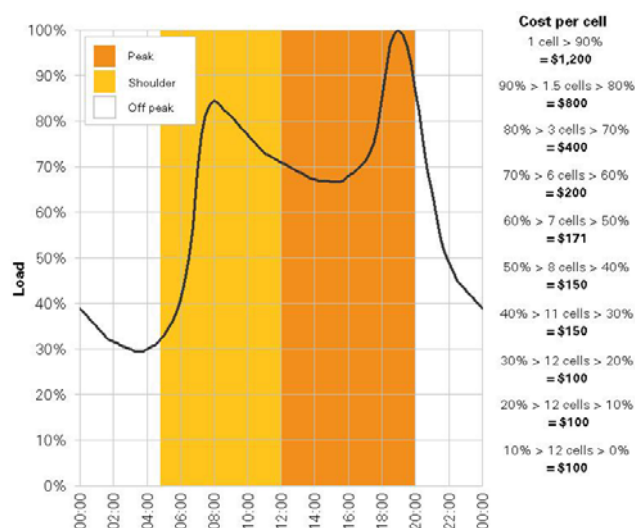
To reflect the relative cost of delivery of network capacity at different times of the day, EnergyAustralia employs a cost

⁸ It should be noted that there are significant costs driven by other factors. For example, the aging of transformers (and therefore the need to replace them) is basically driven by the total energy that passes through it over its lifetime

4. Cost Allocation (continued)

allocation method known as the Method of Intercepts. This method is illustrated in Figure 6, here using a typical winter day profile.

Figure 7 Method of intercepts



The method of intercepts balances the need to allocate sunk costs between customers, but at the same time signal forward costs that are driven by peak use. In this example imagine that \$12,000 must be recovered, and that we wish to allocate this over various times of the day. In this example, we have split the load curve up into 10 rows, and then allocated \$1200 to each row. Then each row is broken up into cells, each 2 hours wide. The \$1,200 for each row is then equally subdivided up to each of the cells in that row. For the bottom four rows, there are 12 cells, so each cell is allocated \$100 (part cells are added together to form whole cells):

$$\$1,200 \div 12 \text{ bands}$$

$$= \$100 \text{ per 'cell' in that row}$$

This calculation is repeated for each row. For row 5, there are 8 cells, so each cell gets $\$1200/8 = \150 per cell. The last row only has one cell, so that cell receives \$1200.

The figures in each column are summed. It can be seen that the 18.00 to 20.00 time band columns has been allocated:

$$\begin{aligned} & \$100 + \$100 + \$100 + \$109 + \$150 + \$171 + \$200 + \$400 \\ & + \$800 + \$1200 \\ & = \$3,330 \end{aligned}$$

This is the total cost to supply capacity during that period. This dollar value can be divided by the total demand in kW in the time band, and by 2 (hours) to create a price per kWh.

If price bands such as peak, shoulder and off-peak are to be set, the figures for the time intervals within each pricing period are averaged. A three level price line has been superimposed on the graph for quick reference.

Modified method of intercepts

The basic Method of Intercepts (Mol) formula was developed in the context of vertically integrated distribution and retail businesses and continues to serve those businesses well.

However, it is unsuited for the present circumstances of a stand-alone network business. About 80% of network costs are associated with the capital expenditure program. That is, the LRMC of network expansion is around 80% of the total cost. But the network expansion is only driven by high loads – typically the top 20% or so. Thus, the cost allocation process has been top weighted, as shown below.

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Figure 8 Method of intercepts for a stand-alone network distribution business



This is the same example demand curve used before. The result of this change has been a much greater emphasis on peak consumption so that customer classes that make use of the network in peak periods would bear a greater share of the costs. EnergyAustralia carried out this exercise using system loading from the financial year 2007-08, and derived the following results:

Table 6 Modified Method of Intercepts Allocation Using 07-08 System Load Profile

Period	Definition	Allocation of Costs
Peak	2pm – 8pm on working weekdays	58.1 %
Shoulder	7am – 2pm and 8pm – 10pm on working week days	33.6 %
Off Peak	All other times	8.3 %

These figures have been used when assessing a tariffs share of costs against each cost pool. Those with peakier loads therefore receive a greater proportion of costs as a result.

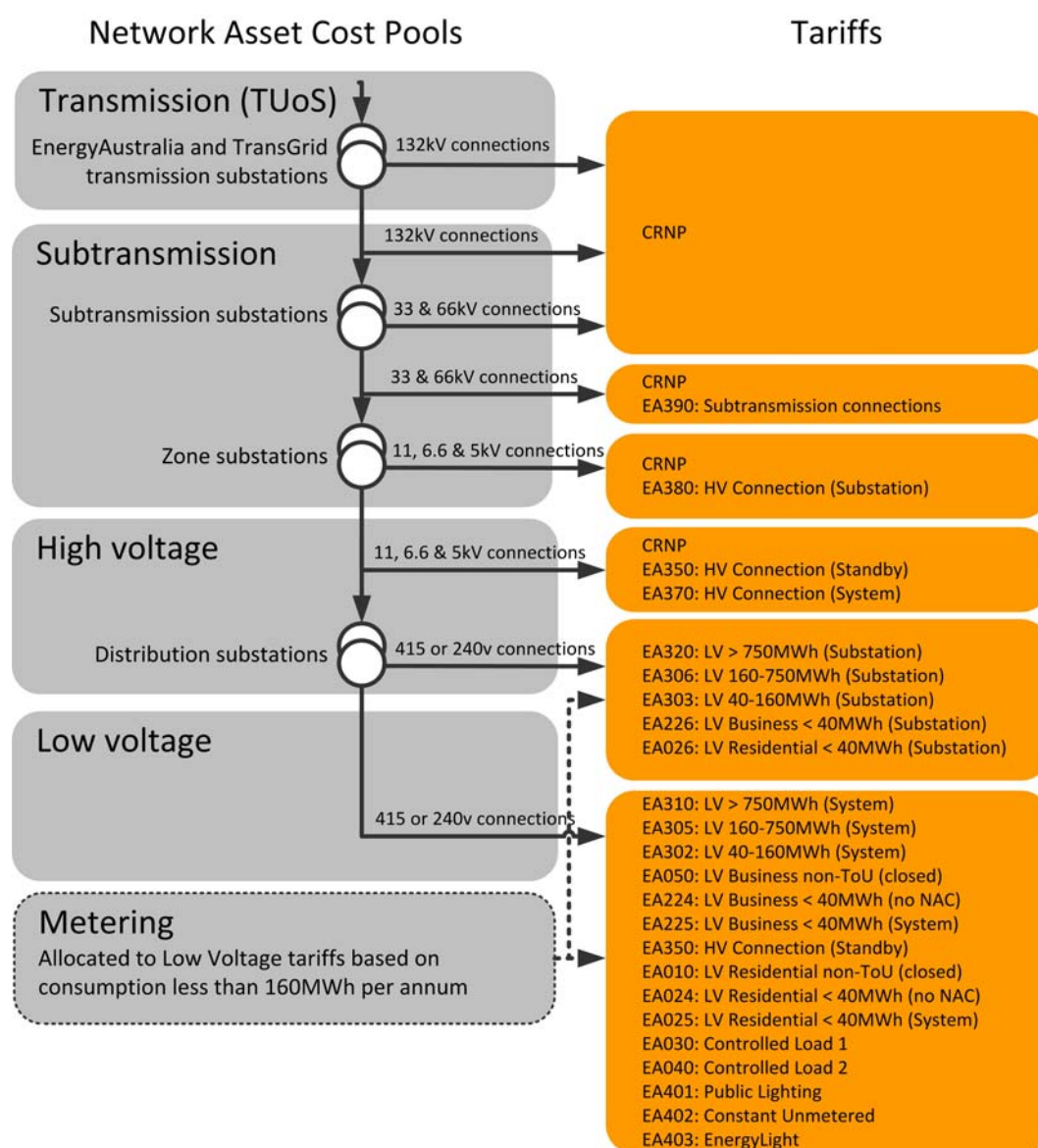
The use of the modified method of intercepts remains unchanged from the 2004-09 regulatory period and is consistent with the distribution price setting obligations under the National electricity Rules.

4.2 Network cost allocation

For costing and pricing purposes, all network asset costs are divided into asset cost pools based on voltage and connection type. Each asset cost pool is allocated the costs associated with providing and maintaining its associated assets. Customers are charged according to the asset cost pools they utilise, which is dependent on their point of connection to the network and their level of peak utilisation. For example, customers who are supplied directly from the high voltage network (mainly 11kV) have prices which do not include the costs of the low voltage network (415 or 240 V). EnergyAustralia's cost pools are presented in Figure X below.

4. Cost Allocation (continued)

Figure 9 Asset cost pool allocation to tariffs



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Within each asset cost pool, costs are allocated using the ‘Method of Intercepts’ process.⁹

The cost pools are valued on the basis of their regulated asset base values. The total revenue stream determined earlier is then attributed to each cost pool in proportion to these values. The regulated asset base values used for 09-10 pricing are current to June 2009. The building block revenue anticipated in the AERs April 2009 determination is \$1,387m¹⁰. The allocation to the cost pool for 09-10 pricing is therefore:

Table 7 Asset Cost Pool Allocation for 2009-10

Asset Cost Pools	Regulatory Asset Base	Revenues attributed to each Asset Cost Pool
EA transmission	N/A ¹¹	\$139.6 ¹²
132kV distribution	\$411.0	\$78.1
132/66/33kV substations	\$453.0	\$86.1
Subtransmission network	\$220.7	\$42.0
Zone substations	\$1,766.5	\$335.8
High Voltage network	\$1,426.0	\$271.1
Distribution substations	\$855.7	\$162.7
LV network and services	\$1,189.8	\$226.4
Metering	\$232.5	\$44.2
Total (excluding transmission)	\$6,555.1	\$1,246.1
Total	N/A	\$1385.7

⁹ Allocating Fixed Costs, Energy International, December 1969.

¹⁰ Includes both transmission and distribution, and accounts for updated CPI of 4.35%

¹¹ Allocated on replacement cost. Refer to EnergyAustralia’s transmission pricing methodology, available at www.energy.com.au/network_prices

¹² Figure based on draft determination as transmission revenue had to be finalised before 30th April 2009

4.3 Changes from building block approach to RAB approach

During the 2004-09 regulatory period, EnergyAustralia employed a building block methodology when assessing the cost allocation for each asset class. A revenue stream for depreciation, return on asset and operational expenditure was determined for each asset cost pool. To simplify the cost allocation methodology, EnergyAustralia has modified the allocation of revenues to each asset class by using the RAB value as discussed above.

The rationale for this departure is that the old method had many assumptions with respect to allocation of operating expenditure in particular. As such it was not immediately clear how the allocation was arrived at. EnergyAustralia has therefore adopted the RAB allocation to simplify the cost allocation. The new approach is easy to understand and explain, and immediately reconciles to the RAB values recognised in the Post Tax Revenue Model used by the AER.

The impacts on the change in approach are reflected in the RAB values by asset class shown below.

Table 8 Cost pool comparison by asset

Asset Category Description	Using building blocks to calculate cost pools		Using RAB to calculate cost pools	
	% Total Revenue	DUoS Revenue	% Total Revenue	DUoS Revenue
132 kV Mains	5.41 %	\$67m	6.27%	\$78m
ST Subs	6.03%	\$75m	6.91%	\$86m
33 & 66 kV Mains	5.38%	\$67m	3.37%	\$42m
Zone Subs	16.92%	\$211m	26.95%	\$336m
HV Mains	17.51%	\$218m	21.75%	\$271m
Dist Subs	14.14%	\$176m	13.05%	\$163m
LV Mains + Services	27.42%	\$342m	18.15%	\$226m
Metering	7.19%	\$90m	3.55%	\$44m
Total		\$1,246m		\$1,246m

4. Cost Allocation (continued)

The impacts of this can be seen in the tariff classes shown below.

Table 9 Cost pool comparison by tariff

Tariff Class	Using building blocks	Using RAB	Variance
CRNP	\$33m	\$32m	-3.0%
Subtransmission Connection	\$5m	\$5m	2.7%
HV Connection	\$30m	\$36m	18.4%
LV Business	\$470m	\$577m	23.0%
Domestic	\$704m	\$589m	-16.3%
Unmetered	\$5m	\$7m	36%
Total	\$1,246m	\$1,246m	0.0%

It can be seen that the zone substations has risen substantially. This reflects the growth in this asset class during the 2004-09 regulatory period. This has reflected in increases to HV tariffs in particular, but it should be noted that this is not particularly as a result in change in method. Rather it genuinely reflects increasing costs in the area of zone substations, as this groups begins a substantial renewal cycle. The other change has been a shift between domestics and non-domestics. Again, these changes are driven more by changes in asset values against each other than by substantial changes in methodology.

4.4 Calculating network use of system prices (NUoS)

EnergyAustralia was required to separate network prices from retail energy prices when the contestable market began in 1996. The original prices were derived from the existing composite retail prices. Since then, EnergyAustralia has made incremental changes to the network use of system (NUoS) prices and price structures in a move towards more cost reflective network pricing.

Prices are calculated for each customer class, or in the case of the largest customers, they are calculated for each customer. As mentioned earlier, network prices consist of three components:

- Distribution Use of System (DUoS)
- Transmission Use of System (TUoS)
- Climate Change Fund cost recovery
- Calculating distribution use of system charges (DUoS)

Cost Reflective Network Price (CRNP) customers

EnergyAustralia calculates individual prices for customers having a demand in excess of 10MW.¹³ This price recovers the costs associated with the customer connecting at a location on the network (so it is also known as a locational price). In overview, the four steps in determining CRNP prices are:

Step 1 – Identify assets utilised in supplying a customer at a load point

Identify the components along the network supply route, in two categories, with separate cost allocation rules:

- Connection assets dedicated to a single customer; and
- The shared network which is used by more than one customer

If \$100 must be recovered from two assets, then each asset is required to recover a part of the \$100 in proportion to its regulatory depreciated replacement cost. So, if one of the assets has a depreciated replacement value of \$5M, and the other a depreciated replacement value of \$10M, then they would receive 1/3 (\$33) and 2/3 (\$67) of the \$100 respectively. The next step is to roll these 'revenues' down to end customers in proportion to their use of those shared assets.

¹³ Eligibility criteria to receive a CRNP (or locational) price are outlined in *ES7 – Application of Network Use of System Charges*, available at www.energy.com.au/network_prices

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Step 2 – Model the network in a pricing simulation

Model the network to the point of connection using simulation software. EnergyAustralia uses a software tool known as TPrice, which is used by transmission businesses to carry out transmission pricing. It analyses the network for each hour of the year and allocates the annual costs of individual assets (the \$33 and \$67 from above) to customers in proportion to their usage of the asset. This model generates an annual charge specific to each large customer for use of the EnergyAustralia distribution network. Transmission costs must also be added to this charge.

Step 3 - Identify customer dedicated asset costs

The costs associated with dedicated assets paid for by EnergyAustralia are applied as a fixed charge or Network Access Charge.

Step 4 – Convert allocated distribution costs to a price

Cost allocations for the shared network are assigned to Peak, Shoulder and Off-peak periods using the Method of Intercepts. These cost allocations are divided by the Time of Use kWh to establish rates based on the consumption in each period. Shared costs are split between energy and capacity charges, based on the principle that the capacity charge reflects reserve capacity on the network. The working assumption is that reserve capacity represents anything above standard N-1 supply, making it effectively N-2. So, a third of variable charges has traditionally been recovered through the capacity charge.

Customers with a demand less than 10 MW

It is only practicable to apply the CRNP methodology to the largest customers. All other customers are grouped in to tariffs and tariff classes on the basis of similar connection, nature and consumption patterns. Thus, each class is charged for NUoS based on the assets it utilises. The offered prices are determined in the following way.

Step 1 - Identify Network costs to be recovered through DUoS Charges

The costs of using the network are divided into the established cost pools. The costs being separately recovered under CRNP (greater than 10MW) are subtracted from this calculation.

Step 2 - Identify customer and demand related costs

Network costs are separated into two components:

- Customer Related - The costs of the low voltage (LV) network are generally proportional to the number of customers, as its construction details are generally independent of the customer size. These customers have the metering, service line and LV construction costs averaged for their use of the network. A portion of these costs is reflected in the Network Access Charge (NAC). Other customer related costs include meter reading and billing; and
- Demand Related – Growth in customer demand drives the need to augment (increase) the capacity of the high voltage and subtransmission networks and their associated substations.

Step 3 – Allocate distribution costs

Each customer is allocated distribution system costs in proportion to the number of customers using the assets and the costs involved. Demand related costs are allocated to customer classes using the Method of Intercepts, which distinguishes between customer classes using their Time of Use energy consumption.

The cost allocation process is illustrated in the next figure. The grey areas indicate that customers in that class are not charged for that portion of the network.

4. Cost Allocation (continued)

Figure 10 Distribution cost allocation

Cost Pool	Network Connections							
	Sub trans	High Voltage		Low Voltage				
		Sub stn	Sys	Sub stn	Sys	Non ToU	Domestic	Street lights
132 kV Network								
132/66 & 132/33 Substn								
Subtrans Networks								
Zone Substn								
HV Network								
Dist. Substn								
LV Network								
Metering connection*								

*Metering only allocated up to 160MWh pa customers

Step 4 – Converting allocated distribution costs to a price

Tariffs are set in each customer class to take into account the requirement to allocate costs to those who incur them (and so remove cross-subsidies) and provide appropriate incentives for demand management.

The following figure shows which distribution cost pools are components of the distribution price. The grey areas indicate no contribution by that class to that charge.

Figure 11 Distribution price structure

Cost Pool	Network charge			
	Capacity	ToU Energy	Anytime Energy	Network Access
132 kV Network				
132/66 & 132/33 Substn				
Subtrans Networks				
Zone Substn				
HV Network				
Dist. Substn				
LV Network				
Metering connection				

Each tariff component has derived in the following manner:

1. Fixed charges are determined by the allocation of the LV network and metering. These are allocated as a proportion of customers. The cents per day rate is then simply the allocated costs divided by the expected total billed customer days for that tariff.
2. All other costs are allocated to variable charges. Where a tariff has only anytime energy (or an inclining block tariff structure), the price is determined by simply taking the costs and dividing by the energy volume. In the case of the inclining block structure, the differential between the first block and second block is based on an assessment of customers capacity to respond to the price signal. Currently the second block is 50% higher than the first block, which creates a meaningful financial incentive for customers to change behaviour. This differential may change over time as EnergyAustralia assesses its effectiveness
3. Where a tariff has a capacity charge as well as ToU energy rates, the allocation between capacity and energy is on a ratio of 33% to capacity and 66% to energy. This is based on the rationale that the capacity charge more or less

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reflects reserve capacity on the network. Networks are generally built with a standard N-1 supply redundancy. Any capacity above this effectively constitutes N-2, or another third of supply, hence an allocation of a third of costs to the capacity charge. This approach in allocation also reflects a balance between allocation of costs from peak use versus allocation from general use in ToU energy rates (this point is discussed in the MOI section above)

Calculating transmission use of system charges (TUoS)

Transmission charges

In NSW, TransGrid sets transmission charges on behalf of all transmission businesses (disclosed in Attachment J). This includes EnergyAustralia's transmission business. In setting those charges, TransGrid requires information from EnergyAustralia about our transmission revenue requirements and how that relates to our transmission assets. Details on this can be found in EnergyAustralia's Transmission Pricing Methodology, available at www.energy.com.au/network_prices. The final charges calculated by TransGrid are set in accordance with the Rules and also guidelines set down by the AER. Transmission prices are set to reflect the cost of the delivery of those services, in particular peak use. Transmission charges faced by EnergyAustralia's distribution business have three components with the following proportions:¹⁴

- Locational fixed charges(14%)
- Locational peak and shoulder energy (29%)
- Locational monthly peak kW demand (29%)
- Non-locational common service and general charges (28%)

When allocating transmission costs to distribution customer classes, the transmission price structures are preserved where possible. The direct mapping of these three components is limited by the customers' metering. The fixed component of

the TransGrid charge is apportioned on the basis of the customers' anytime energy consumption. The mapping is shown in Figure 11.

Where a tariff does not have the relevant structure to immediately pass through the TUoS price component, the allocation is on a second best basis with what is available:

- Demand charges are allocated to capacity components where they exist. Otherwise for TOU only tariffs, through peak and shoulder energy rates, reflecting the time band for which demand charges relate.
- Peak and shoulder energy charges are passed through to peak and shoulder energy rates for TOU tariffs. Otherwise for anytime energy rates, they are a smeared contribution to those energy rates.
- Common services and general charges are allocation to anytime energy rates, as too are fixed transmission charges.

Market settlement surpluses

TransGrid receives settlement surpluses arising from the operation of the National Electricity Market. The rebate, whilst quite variable, is taken into account in TransGrid's published rates. This creates substantial volatility in transmission prices. For this year, because of a substantial decline in forecast settlement residue payments, TransGrid's prices are set to rise by 22% to compensate. Added to this, EnergyAustralia under-recovered in transmission costs in 2008-09 by \$15M. This \$15M shortfall will now be recovered in 2009-10. Combining these two will mean that transmission charges faced by end users will rise by 31% in 2009-10.

Price components

Developing prices that accurately reflect the economic cost of the network service (allocative efficiency) is a key goal in setting prices. This requires the apportioning of costs (cost allocation) to customers as closely as possible to the actual costs incurred in providing the services they use. Prices are then said to be 'cost reflective'. However, cost reflective price levels may be moderated in order to address other goals. The

¹⁴ Forecast for 2009-10

4. Cost Allocation (continued)

goals of simplicity, equity, and managing demand, can run counter to cost reflective pricing and need to be balanced.

Table 10 provides an overview of the major price components. No single pricing component can deliver outcomes that reflect the cost of providing the service. However, by combining two or more components it is possible to get much closer to achieving the desired outcomes. EnergyAustralia Network describes the combination of these components as a price structure. Customers may be placed on more than one price component, especially if they install interval meters (which record usage for each half-hour of the day). In general, EnergyAustralia Network will insist on the most cost reflective price that can be applied, having regard to the capability of the metering

Figure 11 Transmission charges mapped to network prices

EnergyAustralia Network Price Structure	TransGrid Price Component		
	Peak kW Demand, (29%)	Peak and Shoulder Energy (29%)	Fixed + CS & GC (42%)
Capacity Charge	Demand		
Time of Use Energy Charges		Peak and Shoulder energy	
Anytime Energy Charges			Anytime consumption

Table 10 Price Components

Price Component	Purpose or Reason	Comment
Fixed charges	A Network Access Charge applies to all customers for their physical connection to the network.	This recovers the costs of network service that are regarded as fixed and should ideally also recover the excess of average over marginal costs.
Anytime usage charges	Customers are charged according to their total consumption.	These recover variable costs. A single rate does not allow for different usage patterns. Setting different prices in geographic areas would slightly improve cost reflectivity.
Time of use charge	This is an extension of the basic usage charge. Price differentials are applied to consumption at peak, shoulder and off-peak times.	The higher price for peak times can provide a signal to the customer of increased costs associated with usage at this time. Usage at peak times drives network enhancement. Seasonal rates can also provide additional signals, if the sophistication of metering permits.
Capacity charges	These are charges based on the maximum consumption in a half-hour period.	Networks have to cope with the peaks, leaving spare capacity available at other times. It is the peak demand that drives enhancement of the network and incremental network costs. This type of charge targets the customer's contribution to capacity augmentation. A time of use and seasonal loading can also be used to provide additional signalling.

Part 1 – Methodology and Strategy in Setting Network Tariffs

Fixed charges

Fixed charges are the equitable way to recover fixed network costs. In an economic sense, they are also appropriate for recovering the excess of average over marginal costs. Marginal costs should be passed directly on to the customer through the consumption-based price components, to signal the effect of their network usage. Fixed costs are less likely to affect consumption patterns (being independent of usage). In general, fixed charges:

- Cover connection, metering installation & maintenance costs (except those paid for by the customer up-front);
- Cover activities such as meter reading and account processing;
- Cover instrument (voltage and current) transformers used for metering in large low voltage installations and all high voltage installations; and
- Cover cost recovery of the LV network, which only has replacement driven capital ascribed to it on a forward looking basis.

The rationale for recovering the cost of the LV network is on a fixed cost basis. As such it is appropriate to recover the cost of ongoing replacement capital, plus any sunk cost (pre contestable assets) on a least distortionary basis, that being through fixed charges.

Nevertheless, many customers do not like fixed charges or costs that are not directly related to their consumption. This customer preference and the general experience that small consumption domestic customers often have lower incomes has meant that EnergyAustralia has tended to underprice the fixed component of tariffs to small customers. This means some fixed costs are recovered through higher usage rates.

While considering the outcome of this fixed charge allocation, EnergyAustralia is also mindful of setting the fixed charge to make up the shortfall between total average costs and marginal costs. Economically this two part tariff approach allows for efficiently signalling the marginal cost, while making up the shortfall to total costs through the fixed charge in the least distortionary manner. While this is the most efficient outcome, customers still expect the fixed charge to represent specific

costs from a cost allocation. As such, when setting the fixed charge of tariffs, EnergyAustralia must apply business judgement when considering these competing drivers.

Basic usage (consumption) charges

Most deployed meters only provide a total energy consumed over a given time period. They do not provide consumption in time bands or peak demand use. Pricing based solely on the total metered consumption is a poor way of signalling the mainly peak-related network costs to a customer. Customers who use the network in the peak time less than the average for their customer class, would generally face higher charges than if they were on a more cost reflective tariff.

In 2004, EnergyAustralia implemented an Inclining Block Tariff (IBT) structure for all anytime energy use customers. The rationale behind the IBT is that customers that use air-conditioners use significantly more energy than those that do not. Air-conditioning is a primary factor in the recent rise in summer system peaks, which in turn is a major cost driver. The first block is set to cover the consumption of most network users and is grouped according to domestic and business customers. The second block covers consumption above and beyond this level. The IBT structure also provides a financial incentive for large use customers to shift to a more cost reflective ToU tariff.

Time of use (ToU)

Time of use (ToU) consumption charges are possible where meters record energy consumption in specified time intervals. Users are charged more for energy consumption during peak periods, as this generally drives future investment in the network. There is, therefore, an incentive for customers to move non-essential usage to the cheaper off-peak times. Seasonal ToU tariffs are similar to standard daily ToU, but the price varies by time of year in addition to time of day.

Capacity charges

A capacity charge is similar to a demand charge, but differs in its reset period. Whereas the demand charge is based on the peak demand in the current month, the capacity charge is based on the highest monthly peak demand over the preceding 12 months of billing. A capacity charge more adequately

4. Cost Allocation (continued)

apportions a share of the costs associated with network addition to customers with low load factors (who make maximum demands on the system at very infrequent times). Capacity charges are levied on a cents/kW/day (for 40 to 160MWh pa customers) and a cents/kVA/day (for above 160MWh pa customers). This reflects the limitations of metering applied to each of these tranches. Ideally kVA (rather than kW) capacity charges would be charged in all instances.

Outcomes from Cost Allocation

The cost allocation process described above gives a set of potential prices that could be applied to customers. However, other considerations must be included in determining network tariffs. These include principles of simplicity, stability, ensuring revenue security for the network business and in particular whether customer have the capacity to respond to those price signals. While a cost allocation process may well suggest a certain price for peak energy pricing, customers may not be able to respond to those signals. Such an approach may also detrimentally effect overall revenues, limiting the businesses capacity to invest in new capacity.

Renters for example make a good example of people who are unable to make substantial improvements to the thermal efficiency of the house or unit they are renting. This may well impact their ability to reduce their use during peak periods. That being the case, then it is incumbent to consider these issues when setting peak prices. Conversely, peak energy use only makes up about 22% of an average domestic customers total energy use. This means that peak prices need to be relatively high for the dollar impact of peak use to be relatively high on a customers bill. This is important, because it is the total dollar impact of electricity charges that a customers tends to respond to. To that end, the peak charges need to be big enough, such that it makes up a sizeable portion of the total bill. Also, the peak network charge must also compete with retail energy

charges that dilute the network price signal. Taken together, these effects suggest that the peak prices reflected in a particular cost allocation methodology might not be enough to drive meaningful demand response.

As can be seen, it is not just a case of taking the cost allocation prices and applying them in the expectation that they are efficient. The responsiveness of customers is a key consideration. Assessing this responsiveness has been discussed in section 2.4 Assessing the performance of network tariffs.

Cash flows to the business are also an important consideration. As an example, EnergyAustralia was recently considering the introduction of mandated seasonal ToU tariffs . Seasonal ToU tariffs have peak charges that only apply to working weekdays in the summer and winter months, with only shoulder and off peak rates applying for the remainder of the year. This reflects that the network peaks take place in summer and winter, while the loading in autumn and spring is relatively mild. The problem arises in that the general cash flow of the business already reflects this seasonality with income peaking in summer and winter, and dropping off over autumn and spring. The introduction of seasonal prices would amplify this effect on revenues. It can be seen therefore that the introduction of seasonal ToU would create substantial peaks and troughs for the business, creating substantial cash flow difficulties during the non-seasonal periods. This is obviously a consideration when thinking proposed prices that from a cost allocation model.

Summary of Steps in Cost Allocation Methodology

The diagram on the following page summarises the cost allocation methodology described above:

4. Cost Allocation (continued)

4.5 Balancing cost reflective outcomes with the price cap

Having calculated a set of cost reflective rates using the methodology discussed above, the next step is to consider the constraints of the weighted average price cap.

For 2009-10, EnergyAustralia has examined the cost reflective prices and determined to cap increases for this year to 43%. This is based on a rule of thumb of limiting price increases to 15% above the average price increase (being 28% for this year). A 15% limit allows a rough doubling of a price over a five year regulatory period if needed. This balances the need achieve cost reflectivity, while attempting to limit price shocks to customers.

Rebalancing is also carried out to ensure the constraints of the price cap are met. If prices exceed the cap, they are adjusted back to achieve compliance while taking account of the revenue impacts to EnergyAustralia.

Generally, EnergyAustralia has made use of the fact that the side constraints on the DUOS component of tariffs does not apply for the first year of a regulatory period. This has been

done by increasing DUOS charges as a priority over TUOS increases. In subsequent price changes, TUOS charges are proposed to be brought back to cost reflectivity for those tariff classes that are under efficient levels.

A comparison of cost reflective levels versus proposed revenues between DUOS and TUOS is shown in Table 7.

4.5 Conclusion

This part of the proposal has outlined the network price setting process applied by EnergyAustralia for 2009-10. The proposal has also outlined EnergyAustralia's long term network pricing initiatives, particularly in the use of different time based tariffs. These initiatives are aimed at improving the performance of tariffs, to drive efficient use of the electricity network.

This proposal will be updated and submitted annually to the Australian Energy Regulator



Part II – Demonstrating Compliance

1. Outline of compliance with Rules

This second Part of EnergyAustralia's Pricing Proposal assesses all of the requirements set out in Rule 6.18.2 included in Chapter 11 (Transitional Rules) of the National Electricity Rules. Specifically, this part of the proposal sets out the tariff classes, tariffs and charging parameters, and expected revenue for the first regulatory year of the new regulatory control period commencing 1 July 2009 and ending 30 June 2010.

Part II of this pricing proposal is structured as follows:

- Chapter 2 sets out the proposed tariff classes for standard control services, the basis for the proposed tariff classes, and EnergyAustralia's procedures for the assignment and reassignment of customers to tariff classes;
- Chapter 3 sets out the proposed tariffs and charging parameters, and the matters that EnergyAustralia has taken into account when determining these tariffs and charging parameters including the long run marginal costs of providing services to each tariff class, the transaction cost implications and the scope for customers to respond price signals;
- Chapter 4 sets out the weighted average revenue by tariff class and compares it to the efficient revenue boundary defined by the standalone cost and avoidable cost;
- Chapter 5 briefly discusses the variations to tariffs;
- Chapter 6 sets out the obligations on EnergyAustralia to provide revenue under the Climate Change Fund;
- Chapter 7 sets out the approach to passing through the cost of transmission use of system services;
- Chapter 8 sets out the changes since the previous regulatory year, as provided for in the Determination and the Rules; and provides information on how the pricing proposal complies with the requirements for pricing proposals as set out in the Rules.

2. Tariff Classes

RULE REQUIREMENT

Clause 6.18.2(b)(1) of the NER requires that a pricing proposal must set out the tariff classes that are to apply for the relevant regulatory year

In addition, when assigning customers to tariff classes the AER is required to have regard to the following principles;¹⁵

(1) customers should be assigned to tariff classes on the basis of one or more of the following factors:

- (i) the nature and extent of their usage;
- (ii) the nature of their connection to the network;
- (iii) whether remotely-read interval metering or other similar metering technology has been installed at the customer's premises as a result of a regulatory obligation or requirement;

(2) customers with a similar connection and usage profile should be treated on an equal basis;

(3) however, customers with micro-generation facilities should be treated no less favourably than customers without such facilities but with a similar load profile

Clause 6.18.3(d) requires that a tariff class be constituted with regard to the need to group customers together on an economically efficient basis, and the need to avoid unnecessary transactions costs.

Tariff classes are defined in the NER as representing 'a class of customers for one or more direct control services, who are subject to a particular tariff or particular tariffs'.¹⁶ Table 1 defines EnergyAustralia's proposed tariff classes for direct control services. The proposed definitions of tariff classes encompass all customers that are provided with direct control

services by EnergyAustralia, and are therefore required to be assigned to at least one of the specified tariff classes.¹⁷

Tariff classes for Miscellaneous & Monopoly Services are discussed in Attachment E.

Table 1 EnergyAustralia Proposed Tariff Class Definitions

Tariff Class	Definition
Domestic	Applicable to separately metered low voltage (415V or 240V) connections that are private residential dwellings. A residential dwelling is defined as a dwelling where people live on a regular and ongoing basis.
LV Business	Applicable to low voltage (415V or 240V) connections that are used for a non-domestic purpose (domestic being defined as above)
HV Business	Applicable to any connection at high voltage (11kV) that is not otherwise eligible for a CRNP as defined below.
Sub-transmission	Applicable to any connection at a subtransmission voltage (132/66/33kV) that is not otherwise eligible for a CRNP price as defined below.
CRNP	Applicable to connections at any voltage that use more than 10 MW of electricity demand on at least three occasions over a twelve month period.
Unmetered	Applicable to any LV connection that is defined as an unmetered supply by EnergyAustralia in consultation with NEMMCO and in accordance with the clause S7.2.3(Item 5) of the Rules

Clause 6.18.3(d) suggests that in defining the proposed tariff classes, there is merit in allowing customers with different characteristics to be assigned to different tariff classes, and also to ensure that customers with similar characteristics are treated equally. The remainder of this chapter sets out the

¹⁵ Rule 6.18.4(a).

¹⁶ Chapter 10, NER.

¹⁷ This ensures compliance with the requirement set out in clause 6.18.3(b) that each customer for direct control services be a member of one or more tariff classes.

2. Tariff Classes (continued)

reasons for the proposed allocation of customers to tariff classes, in accordance with the requirements in the NERs. In addition, it sets out EnergyAustralia's tariff class assignment and reassignment policy.

Basis for the proposed tariff classes for standard control services

In developing the proposed tariff classes, EnergyAustralia has been mindful of the requirements of clause 6.18.3(d), and has distinguished between tariff classes on the basis of the following customer characteristics, namely:

- the nature of the customer's energy use, and particularly differences in the average daily load profile between domestic and business customers;
- the voltage connection required by the customer, namely low voltage (ie, 415V or 240V), or high voltage (ie, either 11kV or 33kV); and
- the total usage of electricity by the customer (ie, unmetered, metered, usage exceeding 10MW).

The principal reason for using these characteristics as the basis for defining tariff classes is that they have differing affects on the costs imposed on the distribution network in order to provide direct control services. In simple terms, the cost of supplying direct load control services to customers differs according to the shape of the customer's load profile which can be generally ascribed to domestic or business customers, the voltage required, and total electricity use of a customer.

Load shape affects the cost of service provision because the network is built to meet peak load requirements. This means that customers with peakier load profiles that correspond with the network peaks generally require greater network capacity, and therefore higher costs to serve per unit of energy delivered compared to customers with a relatively flat load profile. On average, domestic customers have a morning peak and a larger evening peak, which generally corresponds to the distribution network load peaks and therefore is the primary driver of network augmentation costs of the shared network. In contrast, low voltage business customers tend to have a relatively flat daily load profile and so do not contribute as much to network peaks. Low voltage business customers are generally cheaper to serve as a consequence.

EnergyAustralia is therefore proposing to define tariff classes that distinguish between low voltage business and domestic customers to approximate the known differences in the load profiles between these two groups of customers. In principle, with the proposed rollout of smart metering infrastructure, which will allow interval energy use data to be collected for each customer, such a distinction may no longer be required (as the cost profile is revealed with half hourly data). In the meantime, EnergyAustralia believes that there is merit in defining separate tariff classes for low voltage domestic and business customers.

Separate tariff classes have also been proposed for high voltage business customers and sub-transmission business customers, because of the difference in network costs required to provide services to each of these customer groups.

For example, business customers that connect to the sub-transmission network do not require the 11kV or low voltage networks to facilitate their supply. This is in contrast with business customers who are connected to the low voltage network who require the transmission, sub-transmission, 11kV and low voltage network for their supply. EnergyAustralia considers it appropriate for customers who are connected further up the network to be grouped together in a tariff class. This grouping allows costs to be allocated appropriately.

Similarly, separate tariff classes have been proposed for CRNP customers¹⁸ who are the largest energy users on EnergyAustralia's network, and for unmetered customers because of differences in the network costs associated with the provision of direct control services to these customers.

Cost Reflective Network Price (CRNP) customers are by nature very large electricity users. They use in excess of 10MW capacity. Being large industrial loads, their load profiles are generally flat, and their relative cost to serve is low because of this profile. On the basis of the nature of their use (extremely large flat industrial load) and extent (in excess of 10MW) it is

¹⁸ Customers within the CRNP tariff class use in excess of 10MW of electricity on more than three occasions within a twelve month period.

Part II – Demonstrating Compliance

appropriate to propose that CRNP customers be defined as a separate tariff class.

Unmetered supplies obviously do not have metering, and so it is appropriate to separate this class of customers out so that they do not have metering costs attributed to them. They also have generally flat profiles, making them very cheap to serve as they contribute little to network peaks. This reinforces the need to define them as a separate tariff class.

EnergyAustralia believes that the proposed tariff classes ensure that customers with similar characteristics and therefore a similar cost to serve, are treated on an equal basis. This approach ensures that customers are grouped on an economically efficient basis, and the grouping allows customers with a similar cost base to be considered together for the purposes of ensuring those customers pay a price for delivery of electricity that is within the economically efficient bounds of standalone costs and avoidable costs.

In addition, EnergyAustralia believes that by grouping tariffs within the proposed tariff classes, transactions costs associated with the management of tariff classes by EnergyAustralia, and the costs of transferral of customers between tariff classes can be minimised. This grouping will also ensure that the proposed rollout of smart metering infrastructure, and the scope for customers to adopt time of use tariffs and other innovative tariff products, is not constrained by the costly regulatory requirements that relate to the transferral of customers between tariff classes.

That said, EnergyAustralia is mindful of the need to inform customers when transferring them between tariffs. EnergyAustralia's policy on the transfer of customers between tariffs (as compared to tariff classes) is included as Attachment G to this proposal.

In summary, EnergyAustralia believes that its proposed tariff classes group customers on an economically efficient basis, and this grouping minimises the potential transactions costs associated with managing a large number of tariff classes, in accordance with the rule obligations for tariff classes.

Procedures for the assignment and re-assignment of customers to tariff classes

In accordance with Appendix A of the Final Determination, EnergyAustralia will assign customers to tariff classes using the following procedure:

1. If the customer uses a demand in excess of 10 MW more than three times in a 12 month period, then regardless of the connection voltage, the customer is assigned to the CRNP tariff class.
2. If the customer is a residential supply, then the customer is assigned to the Domestic tariff class.
3. If the supply is unmetered then the customer is assigned to the Unmetered tariff class.
4. If the voltage of the supply is 33 kV, 66 kV or 132 kV then the customer is assigned to the Sub-Transmission tariff class (unless already assigned to the CRNP tariff class).
5. If the voltage of the supply is 5 kV, 11 kV or 22 kV then the customer is assigned to the HV Business tariff class (unless already assigned to the CRNP tariff class).
6. If the voltage of the supply is 415V or 240 V and the customer is not a residential supply, the customer is assigned to the LV Business tariff class.

The criteria for assigning customers to tariff classes has been summarised in Table 2 below.

If a customer's criteria changes, the customer will be re-assigned to the appropriate tariff class. This will be carried out annually consistent with the annual review of customers' assignment to network tariffs (as discussed in Attachment G).

2. Tariff Classes (continued)

Table 2 Criteria for assigning customers to tariff classes

Tariff class	Extent of usage	Nature of usage	Nature of connection to network	
			Metering	Voltage
Unmetered	N/A	Business	Unmetered	415V/240V
Domestic		Domestic	Metered	
LV Business		Business		5kV,11kV,22kV
HV Business				33kV,66kV,132kV
Sub-transmission				Any voltage
CRNP	>10MW			

Customers with micro-generation facilities

It is EnergyAustralia policy to treat customers with micro-generation facilities no less favourably than customers without these facilities but with a similar consumption profile. Allocation of an embedded generation customer to a tariff class will be made on the same basis as other customers; this being the extent and nature of usage, and the nature of the connection to the network. The network tariff will include fixed and variable components and if the customer's demand were to be met entirely by the micro-generator then the levied charge will only be the fixed connection component.

3. Proposed tariffs and charging parameters

RULE REQUIREMENT

Clause 6.18.2(b)(2) requires that the pricing proposal set out the proposed tariffs for each tariff class; and

Clause 6.18.2(b)(3) requires that the pricing proposal set out the charging parameters and the elements of service to which each charging parameter relates, for each proposed tariff.

Rule 6.18.5 sets out the pricing principles that are relevant when determining tariffs and charging parameters. Clause 6.18.5(a) sets out the requirement that total revenue expected to be recovered from tariffs within a tariff class lie within the range defined by the standalone cost of service provision and the avoidable costs.

Rule 6.18.5 further provides that:

(b) A tariff, and if it consists of 2 or more charging parameters, each charging parameter for a tariff class:

(1) must take into account the long run marginal cost for the service or, in the case of a charging parameter, for the element of the service to which the charging parameter relates; and

(2) must be determined having regard to:

(i) transaction costs associated with the tariff or each charging parameter; and

(ii) whether customers of the relevant tariff class are able or likely to respond to price signals.

(c) If, however, as a result of the operation of paragraph (b), the Distribution Network Service Provider may not recover the expected revenue, the provider must adjust its tariffs so as to ensure recovery of expected revenue with minimum distortion to efficient patterns of consumption.

The proposed tariffs and charging parameters are set out for each tariff class in Table 3. In developing the charging parameters, EnergyAustralia has taken account of estimates of long run marginal cost, transactions costs and the scope for customers to respond to price signals. In addition, EnergyAustralia has ensured that its tariffs result in the recovery of expected revenue with the minimum of distortion to efficient patterns of consumption.

This chapter sets out the proposed tariffs and charging parameters for each tariff class for the period of June 2009 to June 2010, and explains the elements of service to which each charging parameter relates. The charging parameters for alternative control services are set out in Attachments B and C. The charging parameters for Miscellaneous and Monopoly Services are in Attachment E.

3. Proposed tariffs and charging parameters (continued)

Table 3 Network tariffs and charging parameters for 2009/10 (excluding GST)

Tariff Class	Tariff #	Tariff Name	Network Access Charge	Network Energy Rates			Step Rates		Daily Capacity Rates	Daily Capacity Rates	
				Non-ToU	Time of Use (ToU)						
			¢/day	¢/kWh	Peak ¢/kWh	Shoulder ¢/kWh	Off Peak ¢/kWh	Step 1 ¢/kWh	Step 2 ¢/kWh	Peak ¢/kW/day	Peak ¢/kVA/day
Domestic	EA010	LV Res non-TOU (Closed)	19.8824					7.4732	10.4895		
	EA024	LV Res <40 MWh (No NAC)			16.8131	3.7525	1.5578				
	EA025	LV Res <40 MWh (System)	27.0482		16.8131	3.7525	1.5578				
	EA026	LV Res <40 MWh (Substation)	19.6459		16.8131	3.7525	1.5578				
	EA030	Controlled Load 1	1.5131	0.6557							
	EA040	Controlled Load 2	5.8381	2.9940							
LV Business	EA050	LV Bus non-TOU (Closed)	59.0957					6.3703	9.5535		
	EA224	LV Bus <40 MWh (No NAC)			17.476	3.8912	1.7942				
	EA225	LV Bus <40 MWh (System)	27.0396		17.476	3.8912	1.7942				
	EA226	LV Bus <40 MWh (Substation)	19.2725		17.476	3.8912	1.7942				
	EA302	LV 40-160 MWh (System)	71.8391		7.6751	5.6507	1.9456			11.6533	
	EA303	LV 40-160 MWh (Substation)	62.1590		7.6751	5.6507	1.9456			11.6533	
	EA305	LV 160-750 MWh (System)	133.3183		8.0026	6.0322	1.6938				11.4966
	EA306	LV 160-750 MWh (Substation)	115.8316		8.0026	6.0322	1.6938				11.4966
	EA310	LV >750 MWh (System)	227.6399		7.0952	5.5794	1.4920				12.6777
	EA320	LV >750 MWh (Substation)	192.9604		7.0952	5.5794	1.4920				12.6777
	EA325	LV Connection (Standby Tariff)	238.2627		7.0952	5.5794	1.4920				
HV Business	EA350	HV Connection (Standby Tariff)	667.1165		5.7768	4.3132	1.0297				
	EA370	HV Connection (System)	635.8363		5.7768	4.3132	1.0297				8.8891
	EA380	HV Connection (Substation)	635.8363		4.8779	3.4606	0.7740				6.0525
Sub-trans.	EA390	ST Connection	732.1019		2.7582	2.1003	0.6651				3.1801
Unmetered	EA401	Public Lighting		5.0842							
	EA402	Constant Unmetered		6.2629							
	EA403	Energy-Light		4.6709							

Part II – Demonstrating Compliance

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There are three elements of service for which charging parameters are defined, namely:

- network access charge per day per connection (¢/connection/day) – this is a flat charge per connection, charged on the basis of the number of days of access to the network for that connection. A connection is defined as any connection to the network requiring a unique National Metering Identifier (NMI) as defined by the NEMMO National Metering Identifier Procedure¹⁹;
- electricity usage charge (¢/kWh) – this is a charge per unit of electricity supplied to a connection point, through a primary tariff or also through an additional secondary tariff (usually controlled load hot water); and
- capacity charge per kilowatt or kiloVoltAmp, per day (¢/kW/day or ¢/kVA/day) – this is a charge based on a customer's maximum demand recorded in any half-hour period between 2pm and 8pm over the twelve months preceding the end of the current billing month. Maximum demand is recorded in either kW or kVA according to the customer's metering functionality.

In addition, there are a number of alternative components for the electricity usage charge including:

- an inclining block charge – this charge which is triggered once a threshold amount of electricity use has been used in each billing quarter. All electricity used after this point is charged at a higher rate based on (¢/kWh). The threshold is 1,750kWh for domestic customers and 2,500 kWh for business customers;
- time of use consumption charges vary according to the time of day of consumption and whether it is a weekday, or weekend/public holiday. There are three charging periods namely:

- off-peak (10pm to 7am all days);
- shoulder (7am to 2pm and 8pm to 10pm working weekdays, 7am to 10pm weekends and public holidays); and
- peak (2pm to 8pm working weekdays).

EnergyAustralia uses the Inclining Block Tariff (IBT) structure for customers with Type 6 metering installations²⁰. The second block rate is 50% more than the first block rate, both for residential and business customers. The block usage threshold is more for business customers than residential customers because business inclining block customers consume more electricity on average than residential customers.

The inclining block pricing structure encourages energy conservation, particularly for customers with air-conditioners who use significantly more energy than average. It also provides a financial incentive for large use customers to shift to a more cost reflective Time of Use tariff.

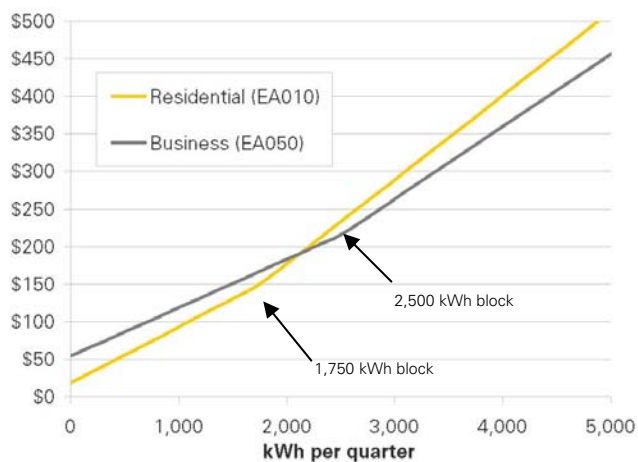
The differential between the first and second block rates of the inclining block tariffs has been set based on an assessment of customers' capacity to respond to the price signal. Currently the second block is 50% higher than the first block, which creates a meaningful financial incentive for customers to change behaviour. This differential may change over time as EnergyAustralia assesses its effectiveness.

¹⁹ NEMMCO National Metering Identifier Procedure 16 July 2006

²⁰ The exception being controlled load, or customers who have migrated to a type 5 (or better) metering installation but have an IBT based retail contract

3. Proposed tariffs and charging parameters (continued)

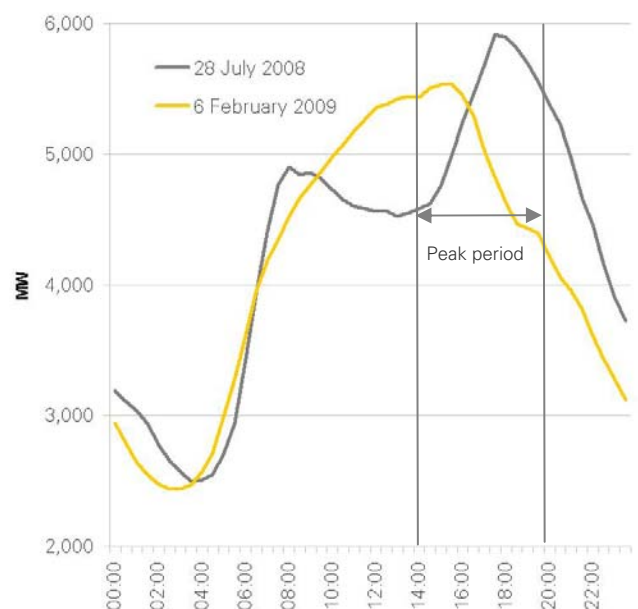
Figure 1 Quarterly network charges for residential and business inclining block tariffs



The time of use periods have been determined to align with the winter and summer peak and off-peak periods across the EnergyAustralia network. Figure 2 shows the demand profiles for the most recent peak summer and winter days.

EnergyAustralia applies a cost allocation methodology to assign costs to each of the charging parameters and thereby determine the appropriate charge. This methodology is summarised in Table 4.

Figure 2 EnergyAustralia summer and winter peak day load profile



Part II – Demonstrating Compliance

Table 4 Cost Allocation Methodology for Each Charging Parameter

Charging parameter		Cost allocation methodology			
Network access charge	DUOS	For LV connected customers, the fixed charge consists of distribution LV mains, services, metering and relay costs. ²¹ For customers connected at higher voltages, there are no DUOS costs allocated to the fixed component, except for CRNP customers, where dedicated assets used to supply those customers is allocated to the fixed charge. For CRNP customers the fixed charge consists of the sub-transmission exit costs, which are determined from an analysis of the existing infrastructure for each customer. 40% of the transmission exit costs are also allocated to the fixed charge (for both tariffs and CRNP customers).			
	TUOS	The cost of dedicated transmission assets used to supply transmission connected CRNP customers is allocated to their fixed charge. there are no other TUoS costs recovered through fixed charges			
Electricity usage charge - inclining block charge	DUOS	These charges reflect costs of the shared upstream distribution network, including distribution substations. Costs are allocated on the calculated peak/shoulder/off peak use of these customers, and weighted against the modified method of intercepts. Consider two tariffs with the following volumes:			
			P/S/OP kWh	MOI Weighting ²²	Sum Product
		Tariff 1	5/2/1 kWh (8kWh total)	58%/34%/8%	3.66
		Tariff 2	2/4/4 kWh (10kWh total)	58%/34%/8%	2.84
		Total			6.50
		If \$100 of asset revenue needs to be allocated between Tariff 1 and Tariff 2, that revenue is apportioned in the ratio of: Tariff 1 = 3.66/6.50 x \$100 = \$56.31 Tariff 2 = 2.84/6.50 x \$100 = \$43.69			
		Having allocated the total dollars to be retrieved in energy charges, the next step is to determine the 1 st and 2 nd block energy rates The relative value of the 2 nd block has been set to be 50% more than the 1 st block. This is based on an assessment of the relative strength of the price signal faced by end users, and their capacity to respond to that price. The rates are set so as to still retrieve the total revenue allocated as described above.			

²¹ Network metering costs are only allocated to customers who consume below 160 MWh pa. Customers above this threshold have meters provided on a contestable basis, so there are no metering costs in their network tariff.

²² MOI refers to the Method of Intercepts allocation described earlier. Refer to Part I of this proposal.

3. Proposed tariffs and charging parameters (continued)

Charging parameter	Cost allocation methodology	
	TUOS	Total TUOS charges payable by the distributor are provided on a forecast basis from TransGrid. These then allocated to tariffs on the following basis: - TUOS exit charges are allocated on a tariffs proportion of peak and shoulder energy - TUOS Common Service and General charges are allocated on the proportion of a tariffs anytime energy against total anytime energy of all tariffs combined. - The forecast revenues from locational charges (both peak and shoulder energy charges together with peak kW demand charges) are totalled, and then allocated in proportion to a tariffs peak and shoulder energy against total peak and shoulder energy of all tariffs combined. Having allocated TUOS charges to the inclining block tariffs, the prices are structured consistent with DUOS prices, with the 2nd block rate being set to be 50% higher than the first block. As mentioned above, this ratio has been set such that the 2nd block rate acts as signal for peak use.
Electricity usage charge - time-of-use charge	DUOS	Having allocated costs using the modified method of intercepts as discussed above (under the discussion for inclining block tariffs), where a tariff has a TOU structure, the peak, shoulder and off peak periods have costs allocated to them using the modified method of intercepts²³ (MOI) as a second round. Extending the example under the inclining block tariff explanation, consider Tariff 1 which has had \$56.31 allocated to it. To determine the costs allocated to peak, shoulder and off peak periods then: Tariff 1 P/S/OP Energy = 5/2/1 kWh, and MOI splits = 58%:34%:8% Then the \$56.31 is allocated in the ratio of $5 \times 58\% : 2 \times 34\% : 1 \times 8\% = 2.9 : 0.68 : 0.08$ So costs to peak/shoulder/off peak = \$44.62 : \$10.46 : \$1.23. These allocated costs are converted to prices by dividing by the forecast energy volumes for each component
	TUOS	TUOS costs are allocated to TOU tariffs consistent with that discussed for the inclining block tariff. The common service and general charges are passed through as a flat rate on top of the peak, shoulder and off peak rates of TOU tariffs. The peak and shoulder energy rates are passed through to the peak and shoulder rates of TOU tariffs. The TUOS kW demand charge is passed through as an equivalent peak and shoulder rate in TOU tariffs.
	DUOS	Where a tariff has a capacity component in addition to TOU energy rates, the costs allocated to between TOU and capacity in the ratio of 67%/33%. This is based on the rationale that the capacity charge reflects reserve capacity above N-1, so they therefore have N-2. This is a third above N-1, hence 33% is allocated to the capacity charge
Capacity charge	TUOS	TransGrid levies locational charges in the form of peak and shoulder energy rates, together with a demand charge. EnergyAustralia receives advice from TransGrid on the forecast total revenues of these components. The ratio of the expected revenues between locational energy and locational demand charges is the basis for how these costs are allocated between peak and shoulder energy rates versus the capacity charge, when the capacity charge exists for a tariff.

²³ Allocating Fixed Costs, H.C.H. Armstead, Energy International, December 1969

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Long run marginal costs

The concept of marginal costs is important in economics because of the general principle that setting prices equal to marginal costs presents consumers with the opportunity costs of their consumption decisions, and results in an efficient allocation of society's resources. The timeframe over which marginal costs are considered to vary with consumption is the distinguishing difference between the concepts of short-run marginal cost (SRMC) and long-run marginal cost (LRMC). In the short-run only costs that directly vary with demand can be avoided (such as the need for some repairs and maintenance, or call centre costs), while in the long-run capital costs can be avoided particularly if anticipated demand growth does not eventuate. The LRMC is equal to the SRMC plus the marginal capital costs

The use of LRMC as a basis for determining usage charges is common in network businesses to provide customers with signals about the incremental capital and operating costs associated with use of the network. The use of LRMC is preferred over SRMC because of a desire to minimise the scope for constraints in the network to ration use of the network. In principle, the use of LRMC ensures that sufficient network capacity is made available such that demand is always able to be met by supply capacity.

Importantly, LRMC is a forward looking concept and should be estimated taking into account future expectations of the growth in costs to meet expected increases in demand. There are a number of methods commonly used to estimate the LRMC for network businesses including the Average Incremental Cost (AIC) approach and the Turvey methodology.²⁴ These two approaches differ in the methodology used to estimate the incremental impact of changes in demand on capacity related expenditure. The AIC approach calculates the average of expected capacity related expenditure over the period that output is expected to change. In contrast the Turvey approach directly estimates the change

in expenditure resulting from an increment or decrement in forecast demand.

While there are strong theoretical underpinnings in favour of the Turvey approach, EnergyAustralia has instead estimated the LRMC for each of the proposed tariff classes using the AIC approach. That said it intends to investigate the appropriateness of alternative approaches to estimating LRMC for tariff classes for future pricing proposals.

The AIC approach can be expressed algebraically as follows:

$$LRMC(AIC) = \frac{PV(capex) + PV(opex)}{PV(incremental_demand)}$$

where the optimal capital costs are the annualised capital expenditure required to meet the additional demand forecast over the period and the operating costs are those required to serve additional demand over the forecast period. The additional demand served is the forecast change in demand compared to a base year. EnergyAustralia considers that incremental demand (rather than energy consumption) is the most appropriate volume cost driver for the LRMC analysis.

The Determination sets out EnergyAustralia's optimal capital expenditure program to meet its overall demand forecasts. This information has been used as the basis for determining the optimal capital costs to be allocated to each of the 2010 tariffs. Similarly the additional demand served has also been allocated to these tariffs. The resulting estimates of the LRMC were aggregated by tariff class and set out in Table 5. These are compared against the weighted average usage charges for each tariff within the tariff class.

Table 5 Estimated long run marginal cost by tariff class, 2010

Tariff Class	LRMC \$ per KVA	2010 DUOS \$ per KVA
Unmetered	\$66.76	\$150.09
Domestic	\$140.87	\$99.38
LV Business	\$96.88	\$84.53
HV Business	\$68.85	\$95.45
Sub-transmission	\$28.05	\$23.24
CRNP	\$32.77	\$63.64

²⁴ See Turvey, R., (1971), *Economic Analysis and Public Enterprise*, Allen and Unwin, London.

3. Proposed tariffs and charging parameters (continued)

The analysis shows that when compared to the proposed 2010 tariffs, the LV residential and business tariff classes are below the theoretical network LRMC. The unmetered, HV Business, and CRNP tariff classes appear to be charged at levels higher than the LRMC. These results occur for CRNPs and for unmetered customers because the incremental volume forecasts for these services increase more quickly than the incremental volumes for the LV residential and business services. The HV Business tariff class has 2010 revenue that is higher than the theoretical LRMC because this tariff class has a higher proportion of network cost attributed to it, without the corresponding higher total volumes.

Transactions costs

In developing the tariffs and charging parameters, EnergyAustralia has been mindful of minimising the transactions costs associated with levying charges. Its decision to charge on the basis of a combination of fixed, usage and capacity charges reflects the trade off between applying a complex set of charges (eg, introducing a power factor charge or additional inclining block rates) and ensuring that charges can be closely aligned to the underlying marginal costs of providing direct control services. Introducing a greater number of tariff and charging parameters increases pricing complexity and will inevitably increase the transactions costs involved in charging for network services, both to EnergyAustralia through the need for more complex (and expensive) customer management and billing systems, and to customers, through additional effort required to understand the basis upon which charges are calculated.

EnergyAustralia believes that its tariffs and charging parameters strike the right balance between transactions costs and ensuring that appropriate signals are provided to customers to facilitate the efficient use of network services.

Scope to respond to price signals

EnergyAustralia's network tariffs and charging parameters provide both short term and long term incentives to customers to modify consumption patterns in line with efficient use of network services. These price signals to consumers are achieved via the design of the energy usage charges and capacity charges, and have been designed to encourage

customers to either reduce or switch their electricity usage. Short term changes to consumption can include moving electricity usage out of the time-of-use peak periods. Long term changes typically occur with installation of, more effective insulation for housing, more energy efficient plant and equipment, or the installation of power factor correction devices e.g. capacitor banks.

EnergyAustralia has used findings from its analysis of customer response to ToU tariffs and the subsequent review by Charles River & Associates (CRA) to determine the likely impact of tariff based initiatives on peak demand and energy growth. EnergyAustralia found an average 1.1 percent drop in demand coincident to the summer system peak for customers subject to ToU pricing initiatives, compared to those who were not.

In its January 2009 revised proposal, EnergyAustralia removed the impact of tariff based demand management from its capital expenditure forecasts. This was on the assumption that EnergyAustralia would not be able to move customers between tariff classes for the purposes of introducing dynamic tariffs that encourage demand management. However, the AER confirmed in its final determination that its revised procedures for assigning and re-assigning customers to tariff classes does not restrict the circumstances in which a reassignment can take place. EnergyAustralia has since re-instated the \$27.9 million capital cost saving to its 2009 to 2014 forecasts.

EnergyAustralia has undertaken a trial to test residential and business customer demand response to new tariff initiatives. The Strategic Pricing Study (SPS) was a two year study that was completed in August 2008. 1,300 EnergyAustralia retail customers were invited to participate in the trial, and interval meters were installed so that half hourly consumption readings could be measured.

During the trial customers were notified in advance of an "event" during which they were charged a very high price for electricity used over the period of the event, usually only 3 to 4 hours. The events were timed to match the predicted maximum demand times on EnergyAustralia's network. When an event was not in progress (ie, most of the time) the participating customers were charged a lower than average rate for electricity.

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The results of the trial were encouraging. Over the 13 events that were called during the two year period, domestic customers showed energy reductions of up to 30% (winter) and 36% (summer), when compared to the control group.

EnergyAustralia intends to continue to explore the potential of such tariffs during the 2009-14 regulatory period. We intend to develop a commercial offering if such tariffs can be demonstrated to be an effective demand management tool.

4. Weighted average revenue

RULE REQUIREMENT

Clause 6.18.2(b)(4) of the NER requires that a pricing proposal set out, for each tariff class related to standard control services, the expected weighted average revenue for the relevant regulatory year and also for the current regulatory year.

Clause 6.18.5(a) provides that:

(a) For each tariff class, the revenue expected to be recovered should lie on or between:

- (1) an upper bound representing the stand alone cost of serving the customers who belong to that class; and
- (2) a lower bound representing the avoidable cost of not serving those customers.

This principle ensures that for each tariff class, the expected revenue is within the bounds that economic principles suggest are consistent with efficient charging practices.

In addition, clause 6.18.5(c) provides that:

(c) If, however, as a result of the operation of paragraph (b), the Distribution Network Service Provider may not recover the expected revenue, the provider must adjust its tariffs so as to ensure recovery of expected revenue with minimum distortion to efficient patterns of consumption.

This chapter sets out the expected weighted average revenue from tariffs within each tariff class for standard control services.²⁵ In addition, it compares these revenues with the standalone costs and avoidable costs by tariff class.

The weighted average revenue for 2009 and 2010, both including and excluding government levies, is set out in Table 6.

Table 6 Weighted Average Revenue by tariff class (\$ million excluding GST)

Tariff Class	Including levies Year ending June 2009	Excluding levies Year ending June 2009	Including levies Year ending June 2010	Excluding levies Year ending June 2010
Domestic	\$643.87	\$639.29	\$746.40	\$729.57
LV Business	\$585.78	\$573.39	\$805.78	\$760.94
HV Business	\$42.13	\$41.21	\$53.96	\$48.94
Sub-transmission	\$13.39	\$13.39	\$14.89	\$14.89
CRNP	\$52.85	\$52.85	\$68.92	\$68.92
Unmetered	\$8.27	\$8.10	\$10.07	\$9.41

Estimating standalone costs

The standalone costs represent the costs that would be incurred to replicate or bypass the infrastructure used to provide a service. It represents an upper bound of costs that should be recovered from customers. If customers were paying above the standalone costs then it follows that it would be economically beneficial for customers to switch to an alternative provider, and similarly if entry was economically feasible it would result in new suppliers entering to provide services. This creates the possibility of inefficient bypass of the infrastructure.

To estimate the standalone costs for a tariff class, EnergyAustralia has used the construct of a hypothetical new entrant distribution network service provider that is seeking to supply services to each tariff class separately. EnergyAustralia has then asked what infrastructure and operating costs would be incurred to supply the tariff class alone, without the benefit of any economies of scale and scope arising from the supply of services to all other tariff classes. To assist with estimating the cost of the hypothetical new entrant, EnergyAustralia has drawn upon information from its own costs as follows:

First, the segments of the network that would be required to provide services to each tariff class have been identified. For example, for HV Business customers, the cost of replicating the network from the point of customer connection to the high voltage network to the connection with the TransGrid transmission network was identified.

²⁵ The weighted average revenue for miscellaneous and monopoly services is set out in attachment E.

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Second, the annual capital charge for the relevant network was estimated applying the same return to capital and depreciation parameters as set out in the Determination.

Third, the operating costs that would be incurred in order to provide services to the tariff class were estimated by identifying those operating costs that are necessarily incurred by EnergyAustralia for the provision of network services to that tariff class.

The associated summation of the annual estimated capital costs and operating costs for each tariff class are therefore estimated to represent the standalone costs that would be incurred by a hypothetical new entrant. The estimated standalone costs for each tariff class are set out in Table 7.

Table 7 Stand-alone costs compared to 2010 network tariffs (DUOS component only)

Tariff Class	Total standalone costs 2010 (\$m)	Weighted average revenue 2010 (\$m)	Weighted average revenue as a proportion of standalone costs (%)
Unmetered	\$1,082	\$7.3	1%
LV Business	\$1,155	\$577.2	50%
Domestic	\$1,173	\$589.1	50%
HV Business	\$774	\$35.5	5%
Sub-trans.	\$580	\$5.0	1%
CRNP	\$776	\$31.9	4%

It can be seen that costs need to establish and maintain a standalone service for each tariff class is significantly higher than the weighted average revenue to be recovered in 2010.

Estimating avoidable costs

The avoidable costs represent those costs that could be avoided by a business if it was not supplying goods or services to its customers. It represents the lower bound of costs that should be recovered from customers. If customers were charged below the avoidable costs it would be economically beneficial for the business to stop supplying the customer and thereby avoid the associated costs which would exceed the revenue expected to be obtained from the customer.

To estimate the avoidable costs for each of the proposed tariff classes, EnergyAustralia has:

- identified those categories of costs that would be avoided if a tariff class was no longer served;
- allocated the cost of each avoidable cost category to each tariff class based on either volume or customer numbers, according to the nature of the cost category, to determine those costs that would be reduced if a particular tariff class was no longer served;
- summed the allocated avoidable cost for all avoidable cost categories for each tariff class to estimate the total avoidable cost for each tariff class.

The cost categories that were identified as being avoidable included:

- repairs and maintenance – this was identified as a partially avoidable cost given that use of the network impacts on its deterioration and the need for repairs and maintenance;
- customer service – these costs relate to managing billing and customer service enquiries which are related to the number of customers served;
- metering costs – these costs relate to meter reading and meter replacement and are related to the number of customers served;
- corporate and divisional support costs – costs relating to media, marketing and legal expenses can be considered to be driven by the number of customers served; and
- customer connections and installation inspections – these costs can be considered to be proportional to customers numbers, the voltage and capacity of the connection

Table 8 sets out the avoidable costs for each of the tariff classes.

4. Weighted average revenue (continued)

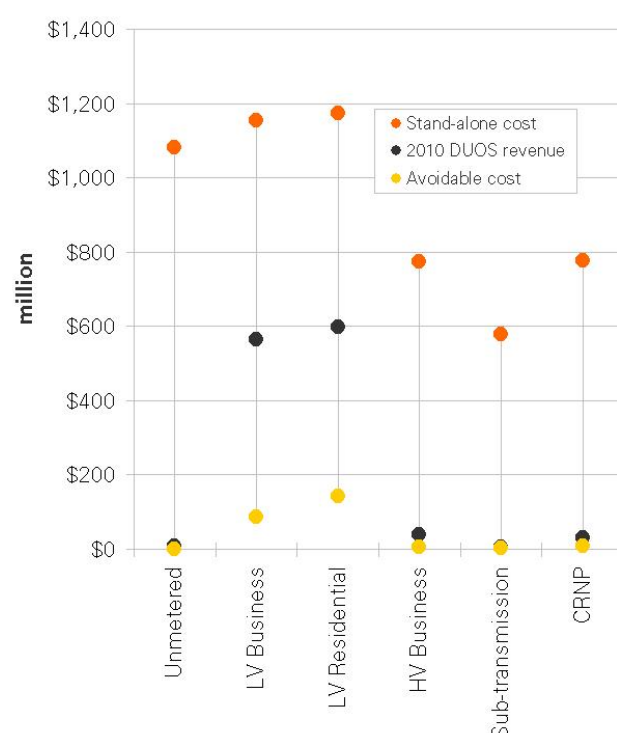
Table 8 Avoidable costs compared to 2010 network tariffs (DUOS component only)

Tariff Class	Total avoidable costs 2010 (\$m)	Weighted average revenue 2010 (\$m)	Avoidable costs as a proportion of weighted average revenue (%)
Unmetered	\$0.72	\$7.3	10%
LV Business	\$87.25	\$577.2	15%
Domestic	\$141.61	\$589.1	24%
HV Business	\$5.20	\$35.5	15%
Sub-trans.	\$3.04	\$5.1	60%
CRNP	\$6.99	\$31.9	22%

It can be seen that in most cases the revenue to be recovered for each tariff class is well above the avoidable cost for each tariff class. The sub-transmission tariff class has a lower weighted average revenue (per unit) when compared to the other tariff classes. This is because sub-transmission network tariffs are lower when compared to the other tariff classes. Hence the avoidable costs for sub-transmission services make up a higher proportion of the equivalent 2010 revenue.

Figure 3 compares the 2010 weighted average revenue with the estimates of standalone and avoidable costs for each tariff class. The revenue falls within the bounds of the upper limit (standalone cost) and the lower limit (avoidable cost).

Figure 3 Efficient bounds for each tariff class



Recovery of Total Regulated Revenues

The application of the WAPC form of price control creates circumstances where EnergyAustralia would not be able to recover its total regulated revenues. This section briefly describes the WAPC formula and the approach adopted to estimate consumption to be used in the formula when new tariffs are introduced, or customers move between tariffs.

EnergyAustralia applies a weighted average price cap (WAPC) form of price control that allows for individual tariffs to be adjusted, while limiting the annual percentage change in the weighted average of all tariffs.

The weighted average price cap formula is set out below:

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$$\frac{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^t * q_{ij}^{t-2}}{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{t-1} * q_{ij}^{t-2}} \leq (1 + \Delta CPI) * (1 - X_t) * (1 + D_t)$$

i=1,...,n and j=1,...,m

where:

- n is the number of distribution tariffs
- m is the number of distribution tariff components

p_{ij}^t is the proposed price for component j of the distribution tariff i for Year t

p_{ij}^{t-1} is the price charged by the DNSP for aggregate component j of the distribution tariff i in Year t-1 (being the Year immediately preceding Year t)

q_{ij}^{t-2} is the audited quantity of aggregate component j of the distribution tariff i that was charged by the DNSP in Year t-2 (being the Year immediately preceding Year t-1)

X_t is the value of “X” specified for the DNSP (for the Year t) by the AER

ΔCPI is the December to December change in the Consumer Price Index from Year t-2 to Year t-1

An important feature of the WAPC formula is the use of lagged electricity consumption data to determine current year prices. Lagged data is used because of the delays in this data being audited and made available, and the difficulty of using forecast data within the formula.

The use of lagged consumption data impacts on EnergyAustralia’s ability to recover its revenue requirement in circumstances when actual consumption in a year varies considerably from lagged consumption. This can occur when new tariffs or charging parameters are introduced, or when customers move between tariffs.

In these circumstances, EnergyAustralia applies a ‘reasonable estimates’ methodology to calculate the volume to be included in the WAPC formula. This ensures that revenue is maintained despite the tariff changes. For each year of the 2009 to 2014 distribution determination, EnergyAustralia will submit reasonable estimates of consumption to the AER for use in the WAPC formula if appropriate. These estimates are based on the quantities that would have been sold in year t-2 if the new tariffs or tariff components had been introduced in that same year, or if EnergyAustralia mandated the transfer of customers from one tariff to another. Voluntary tariff transfers are not included in reasonable estimates unless they relate to the creation of a new tariff.

Table 9 WAPC Notional Revenues (\$m)

	Year t, 2010	Year t-1, 2009
total Non-DUOS Tariffs	13	11
total DUOS Tariffs	1,230	1,001
Combined standard direct control	1,244	1,011
Change in weighted average prices	22.99%	

5. Variations to Tariffs

RULE REQUIREMENT

Clause 6.18.2(b)(5) of the NER requires that a pricing proposal set out the nature of any variation or adjustment to the tariff that could occur during the course of the regulatory year and the basis on which it could occur

EnergyAustralia does not anticipate any changes to its tariffs during the course of the next regulatory year.

6. Climate Change Fund

RULE REQUIREMENT

Clause 6.18.2(b)(5A) of the NER requires that a pricing proposal must set out the amount paid to the Climate Change Fund in or in respect of the relevant regulatory year and reflect that amount in the expected revenue for the relevant regulatory year

For the year ending June 2010, EnergyAustralia will pay \$67,353,873 to the Climate Change Fund. EnergyAustralia expects to recover \$67,353,873 for this purpose from network charges. Attachment F is the is the correspondence received from the Department of Environment and Climate Change confirming the amount of this levy.

7. Transmission Use of System Services

RULE REQUIREMENT

Clause 6.18.2(b)(6) of the NER requires that a pricing proposal must set out how charges for transmission use of system services are to be passed on to customers and any adjustments to tariffs resulting from over or under recovery of those charges in the previous regulatory year. In addition, clause 6.18.7 states that recovery of transmission revenue should not exceed the estimated amount of these charges for the relevant regulatory year, once the overs and unders account has been taken into account.

EnergyAustralia's network charges recover revenue for its own transmission network and also for the TransGrid transmission network. Each year, EnergyAustralia determines the amount of revenue it should recover for its transmission assets from the post-tax revenue model. This annual revenue requirement is sent to TransGrid who include this information in the model used to allocate revenue across the entire NSW transmission network. This package is known as "TPrice" and uses Jacobian equations to model the load flow of electricity throughout the transmission network. The load flows are then used to distribute the revenue that is to be recovered.

Once TPrice has been run, TransGrid provides a spreadsheet to EnergyAustralia that contains the revenue to be recovered for the TransGrid and EnergyAustralia transmission system from EnergyAustralia's network customers. This revenue is broken down into transmission node identifier TNI, fixed charges, exit charges, usage charges, general and common services charges. It also includes load flow summaries across each TNI.

EnergyAustralia allocates the transmission revenue to the network tariffs and CRNP customers predominantly on the basis of forecast volumes. CRNP customers are allocated their share (by volume) of the transmission charges for their corresponding TNI. The transmission exit charges are allocated on the number of circuit breakers attributable to the CRNP customer, while the TransGrid fixed and usage charges are allocated based on the peak and shoulder volume forecasts of the CRNP customer. The remaining TransGrid transmission recovery is spread across EnergyAustralia's tariff customers on the basis of peak and shoulder volumes for each tariff.

Unders and overs account

The over and unders TUOS account is a mechanism to ensure that if transmission revenue recovery is too high or too low over the regulatory year, transmission operators can make up unrecovered revenue in the subsequent year. Fluctuations in revenue recovery are most likely to be caused by the impact of weather on electricity consumption and demand forecasts. The overs or unders account balance for the prior year is added to the revenue to be recovered in the current year, before being sent to TransGrid for calculation via TPrice.

Unders and overs account for FY 2009

In February 2008, TransGrid advised EnergyAustralia of revenues that were required to be recovered from EnergyAustralia distribution network customers in FY 2009. However, due to changes in the settlement residues expected by TransGrid in FY 2009, TransGrid updated (and increased) its revenue forecast that EnergyAustralia was required to recover from its customers in May 2008. Unfortunately, this occurred after EnergyAustralia had finalised its prices for the 2009 financial year. As a result, and despite targeting a zero transmission balance in FY 2008, EnergyAustralia was required to pay to TransGrid a sum \$15 million more than that which would be recovered through prices set for FY 2008. This has resulted in a \$15 million under-recovery balance in FY 2009 that needs to be recovered in FY 2010.

Unders and overs account for FY 2010

As a result of this incident, EnergyAustralia sought a rule change to ensure that transmission companies advised annual price changes by 15 March each year. Whilst considering the rule change, TransGrid and EnergyAustralia reached an agreement that transmission revenues to be recovered would be notified by 6 April each year allowing EnergyAustralia to meet its obligations to publish approved prices one month prior to those prices coming into effect.

To ensure this arrangement between TransGrid and EnergyAustralia was compliant with the Rules, TransGrid sought a rule change to explicitly allow it to use the AER's draft determination Maximum Allowable Revenue (MAR) to set

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prices for the first year of a regulatory period and recover any shortfall or redistribute any over-recovery in the following year through the normal unders- and overs- mechanism. This Rule change was approved by the AEMC on 26 March 2009.²⁶

EnergyAustralia's and TransGrid's draft transmission revenue decisions were used to calculate the final network prices that will apply for FY 2010 and come into effect on 1 July 2009. This was done to facilitate TransGrid's requirement to provide revenues by 6 April and therefore allow them to be used in the TPrice modelling. As allowed in the Rules, any differences between the final and draft decision transmission revenue recoveries in FY 2010 will be accounted for in the overs and unders account and be recovered or redistributed in FY 2011 prices.

Table 10 Overs and Unders Account forecast closing balance

Financial year ending	t-2 audited 2008	t-1 expected 2009	t forecast 2010
Interest rate applicable to balance %	9.70%	9.70%	8.78%
Opening balance \$ 000	3,256	(5,979)	(14,469)
Interest on opening balance (365 days) %	316	(580)	(1,270)
forecast over / (under) recovery for financial year	(9,118)	(7,552)	15,090
Interest charged on over/under recovery for financial year \$000	(432)	(358)	649
Closing balance of transmission overs/ (unders) account \$000	(5,979)	(14,469)	0

²⁶ AEMC National Electricity Amendment (Transmission Network Prices Publication Date) Rule 2009

8. Changes from the Previous Regulatory Year

RULE REQUIREMENT

Clause 6.18.2(b)(8) of the NER requires that a pricing proposal must describe the nature and extent of change from the previous regulatory year and demonstrate that the changes comply with the Rules and any applicable distribution determination

This chapter sets out the principal changes since the previous regulatory year, and demonstrates that these changes comply with the Rules and the AER's distribution and transmission determinations released on 30 April 2009.

Compliance of tariffs and charging parameters with the Determination

The AER published the final determination (the Determination) for the New South Wales Distribution Network Service Providers on 30 April 2009. The Determination sets out the X-factors to be used by EnergyAustralia to determine the rate of change to the weighted average price cap (WAPC) for each year over the regulatory period. These new X-factors are the principal change since the previous regulatory year and are reproduced in Table 11 below.

Table 11 AER final decision X factors for EnergyAustralia

	FY 2010	FY 2011	FY 2012	FY 2013	FY 2014
Distribution	-17.86%	-12.0%	-12.0%	-12.0%	-8.0%
Transmission	-7.77%	-11.0%	-11.0%	-11.0%	-11.0%

Compliance of tariffs and charging parameters with the Rules

Rule 6.18.6 provides that the weighted average revenue per tariff class for standard control services cannot change by more than two percent from the preceding year, after adjustments for the X-factor and the consumer price index have been applied. In consultation with the AER it was established that this constraint does not apply for the first year of the new regulatory control period, but will be applied to subsequent years. The proposed changes to the charging parameters for 2009/2010 are consistent with the Rules and on average do not exceed the price change limits set by the AER's Determination.

WAPC Compliance Model

To demonstrate compliance with the Determination and the Rules, EnergyAustralia has provided the AER with a compliance spreadsheet that sets out the detailed assumptions used to apply the weighted average price cap (WAPC). These assumptions include:

- usage volumes and customer numbers for the year ending June 2008
- adjustments to usage volumes and customer numbers for reasonable estimates
- Distribution use of system (DUOS), transmission use of system (TUOS) and Climate Change Fund rates by tariff for the year ending June 2010
- network use of system (NUOS) rates by tariff for the year ending June 2010
- revenues by tariff and tariff class for the years ending June 2009 and 2010

The compliance spreadsheet included as Attachment D demonstrates EnergyAustralia's compliance with the Determination and the Rules.

Stand-by tariffs

EnergyAustralia has been using our standard capacity tariffs to charge for standby supplies. However the capacity charge is only initiated when the customer uses the supply. The capacity charge then remains set at that level for 12 months. To avoid this effect on standby supplies, standby specific tariffs have been created to apply to HV connections and LV connections. These two tariffs (with codes EA325 and EA350) consist of a large fixed charge and standard energy rates, and will be available on an application basis. The fixed charge is designed to recover the cost of the feeder, even while no capacity is being used. Whether or not the customer uses the supply, the customer will be charged a fixed daily charge for the provision of the supply. This should simplify billing for these supplies. It will also act as a clear signal of the cost to EnergyAustralia of providing the stand-by supply capacity. This is consistent with our strategy to keep tariffs simple and cost reflective where possible, and to send appropriate pricing signals to our customers.

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Change in Allocation Methodology

In determining the value of each asset cost pool, EnergyAustralia has traditionally carried out a full building block analysis. This has meant that each asset cost pool has its own figure determined for operating expenditure, depreciation and rate on asset. The asset cost pools can then be summed back to the revenue entitlement allowed under the regulatory determination.

For the 2009-14 regulatory period, EnergyAustralia has decided to modify the approach in determining the allocation of revenues to cost pools. The new approach uses the latest asset class RAB values to allocate revenues. In this way, operating expenditure, depreciation and return on assets are all allocated to the asset classes in terms of their relative RAB values.

This approach is simpler in application, and delivers a similar cost allocation outcome to the full building block allocation in any case. This is because the depreciation allowance for an asset class is generally in alignment with an asset classes relative RAB size (the bigger the RAB, the more depreciation there is), while the return on asset is exactly aligned. Also, it acknowledges that operating expenditure is only 40% asset related, the majority relates to non-system operating expenditure which must be spread across all tariffs in any case.

Change in allocation to CRNP Customers

EnergyAustralia has used depreciated replacement cost values in allocating costs for calculating CRNP (site specific) tariffs. The previous approach was to use replacement cost values consistent with the approach used in transmission pricing. Use of depreciated replacement was used to make it consistent with the cost allocation used for standard tariffs, with the same underlying rationale.

Standardisation of Demand and Capacity Charges

EnergyAustralia has removed demand components from a small number of CRNP customers. In its place, capacity charges have been applied. Some CRNP customers also have capacity charges applied for the first time this year. This is a final step in standardising tariff structures to be consistent throughout our tariff list

Changes to customer bills

Table 12 Overall Price Change (including the Climate Change Fund)

Tariff Class	2009 Total NUOS \$m	2010 Total NUOS \$ m	Percentage Increase %
Unmetered	\$8.10	\$10.07	24.3%
LV Business	\$567.75	\$805.78	41.9%
Domestic	\$639.29	\$746.40	16.8%
HV Business	\$41.21	\$53.96	30.9%
Sub-transmission	\$13.39	\$14.90	11.2%
CRNP	\$52.85	\$68.92	30.4%
Total	\$1,322.59	\$1,700.02	28.5%

Figure 4 Total weekly network bill increases for customers on inclining block tariffs (network tariffs EA010 and EA050)

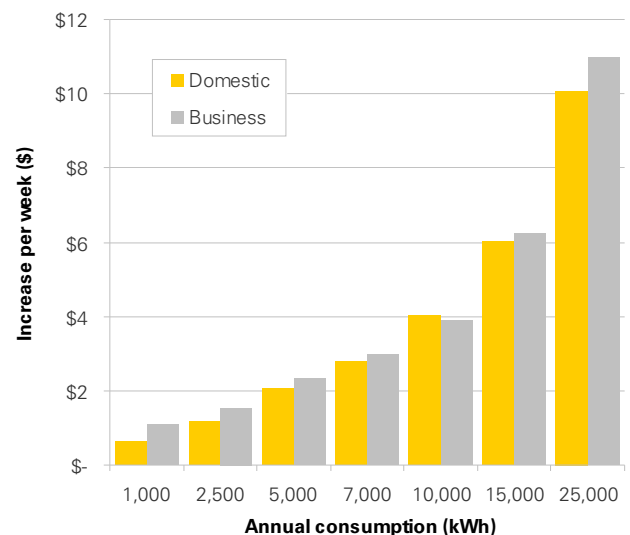


Figure 5 Price change scatter plot for a sample of EA010 customers

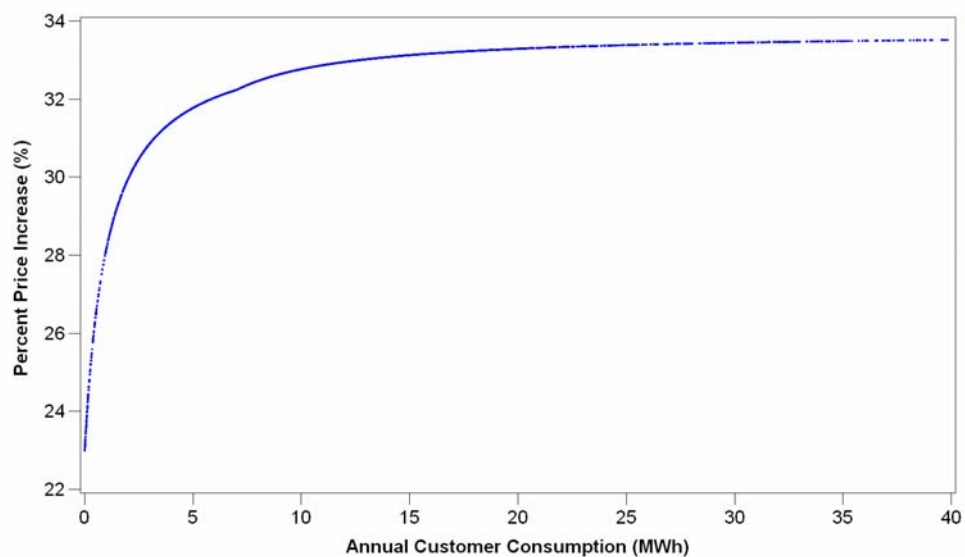
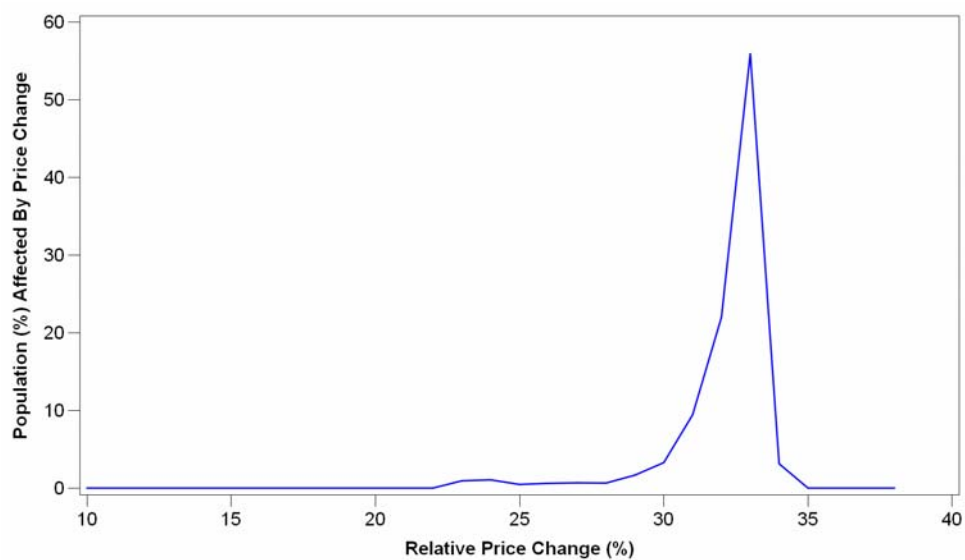


Figure 6 Price change histogram for a sample of EA010 customers



Part II – Demonstrating Compliance

Figure 7 Price change scatter plot for a sample of EA025 customers

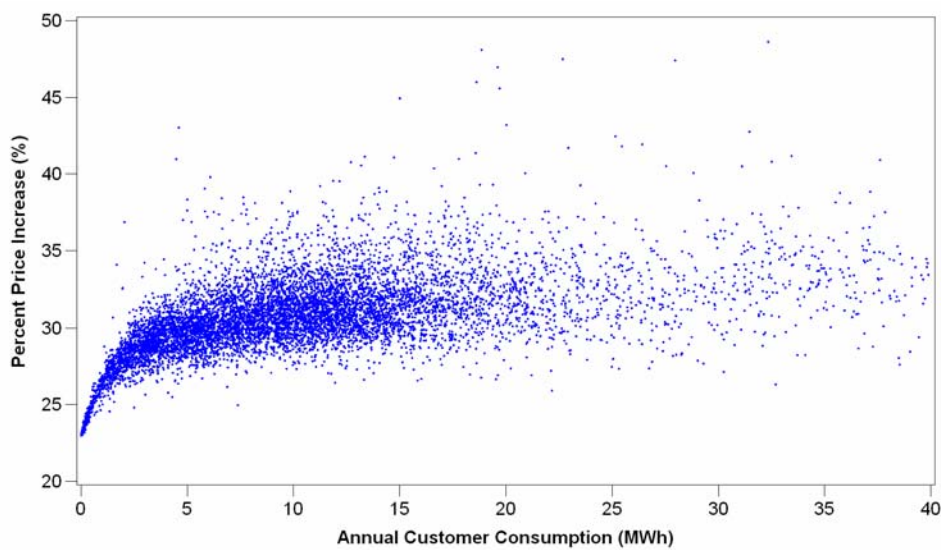
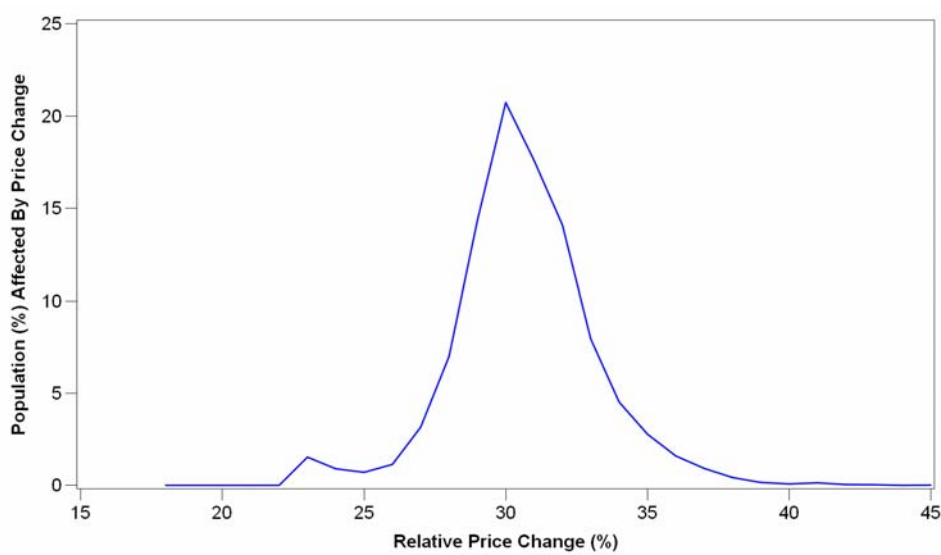


Figure 8 Price change histogram for a sample of EA025 customers



8. Changes from the Previous Regulatory Year (continued)

Figure 9 Price change scatter plot for a sample of EA050 customers

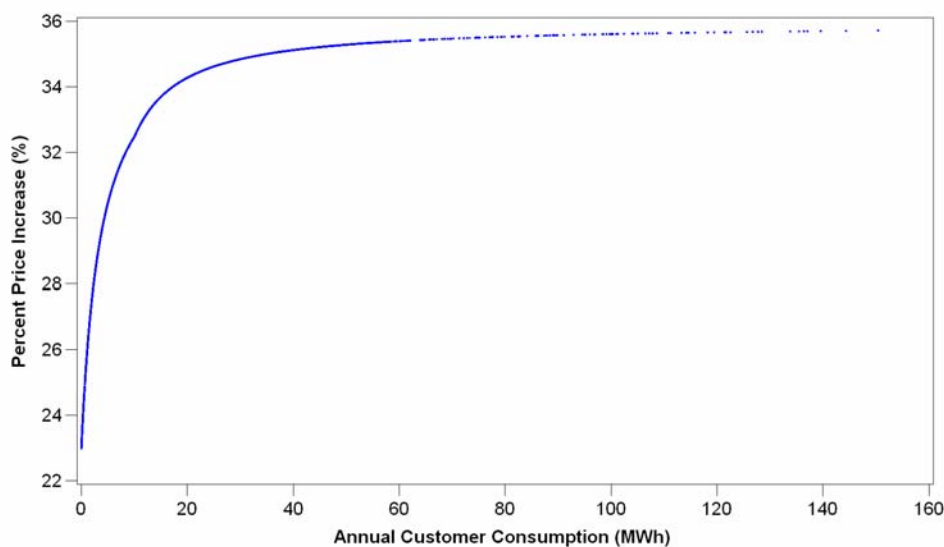
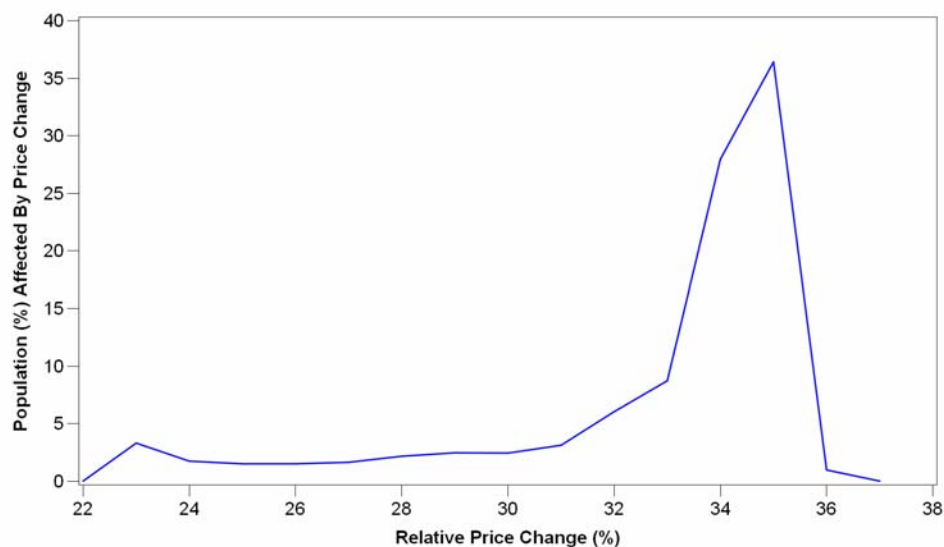


Figure 10 Price change histogram for a sample of EA050 customers



Part II – Demonstrating Compliance

Figure 11 Price change scatter plot for a sample of EA302 customers

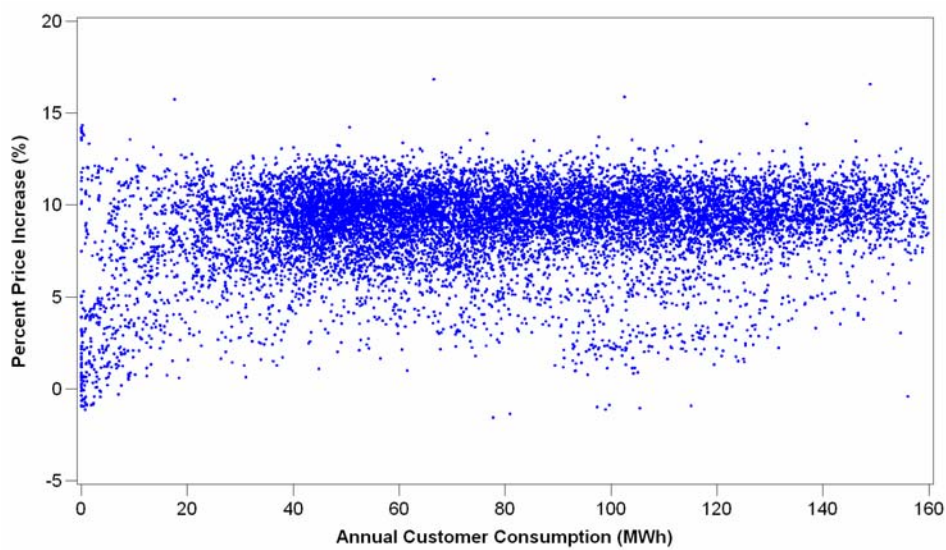
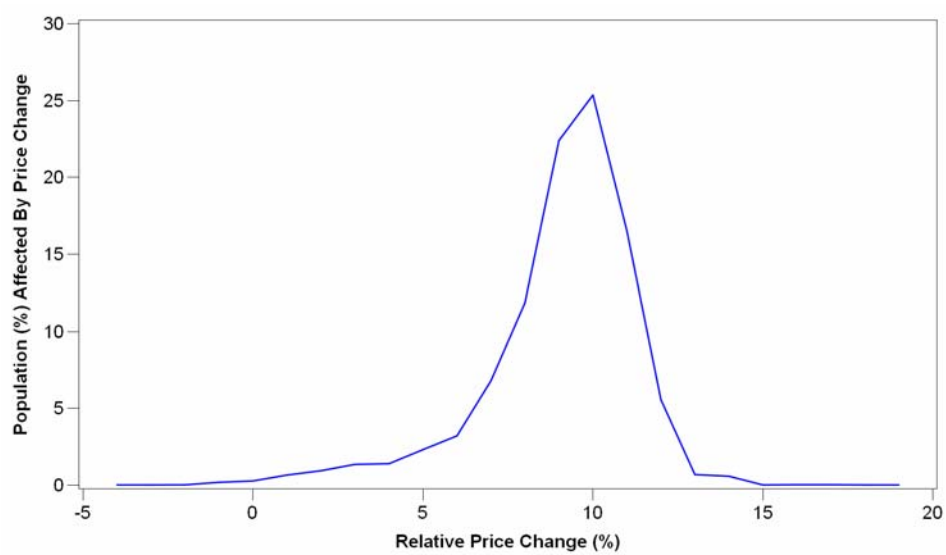


Figure 12 Price change histogram for a sample of EA302 customers



8. Changes from the Previous Regulatory Year (continued)

Figure 13 Price change scatter plot for a sample of EA305 customers

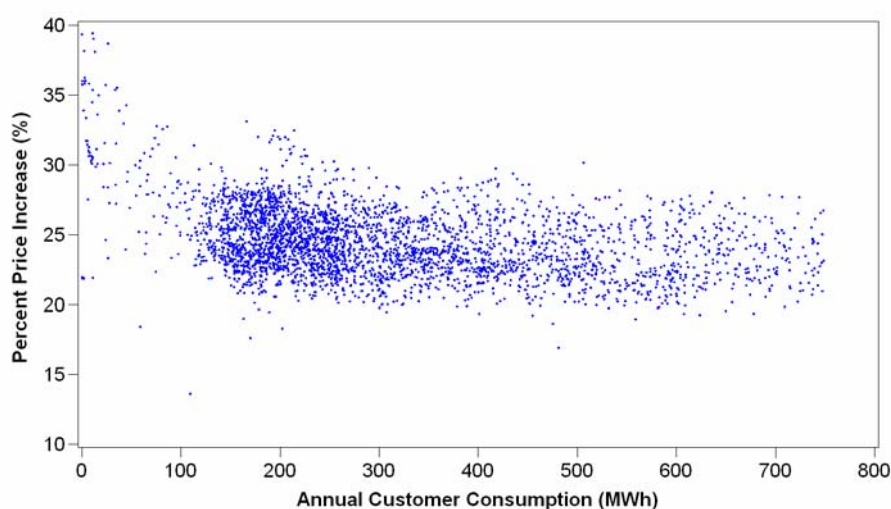
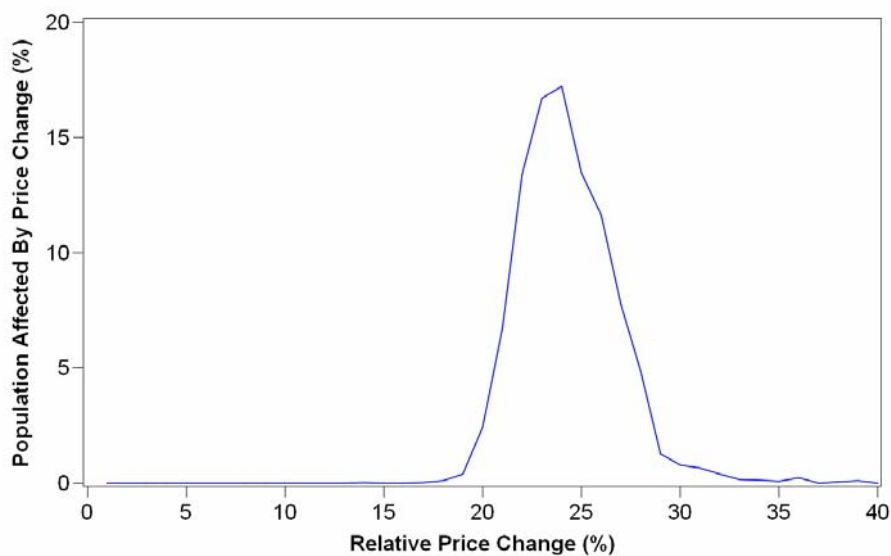


Figure 14 Price change histogram for a sample of EA305 customers



Part II – Demonstrating Compliance

Figure 15 Price change scatter plot for a sample of EA310 customers

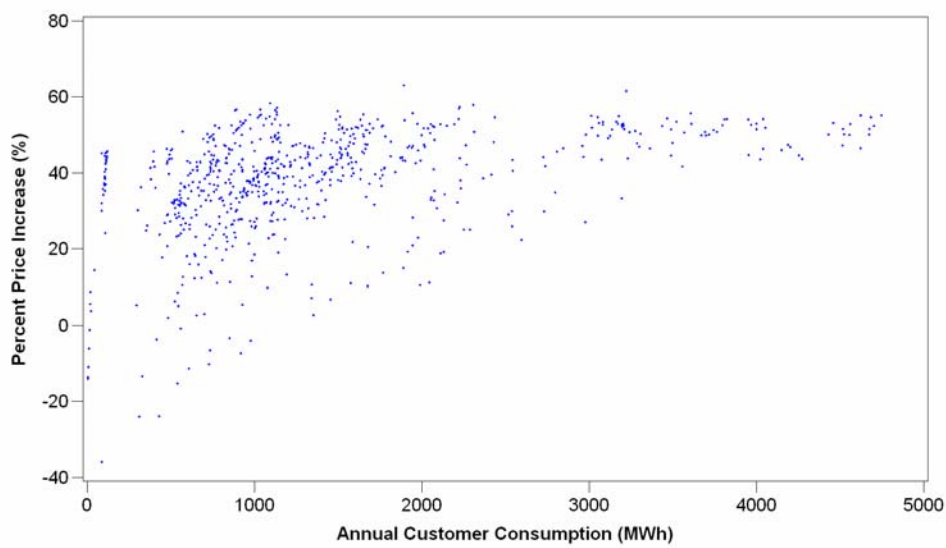
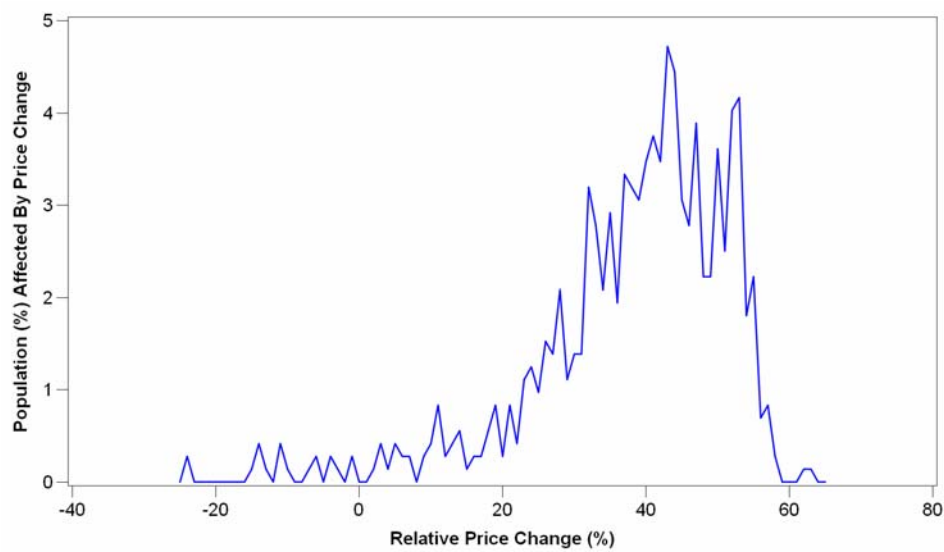


Figure 16 Price change histogram for a sample of EA310 customers



8. Changes from the Previous Regulatory Year (continued)

Figure 17 Price change scatter plot for a sample of EA370 customers

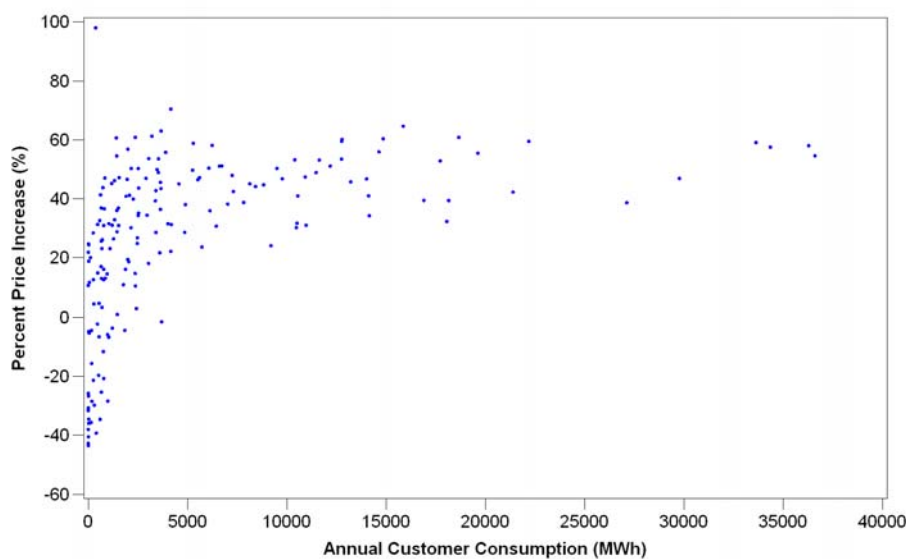
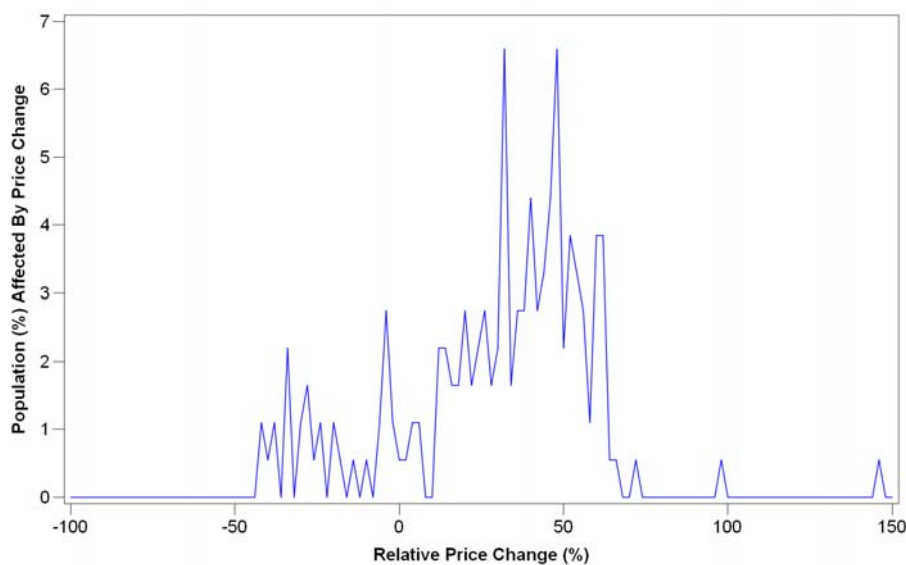


Figure 18 Price change scatter plot for a sample of EA370 customers



Part II – Demonstrating Compliance

Figure 19 Price change scatter plot for a sample of EA390 customers

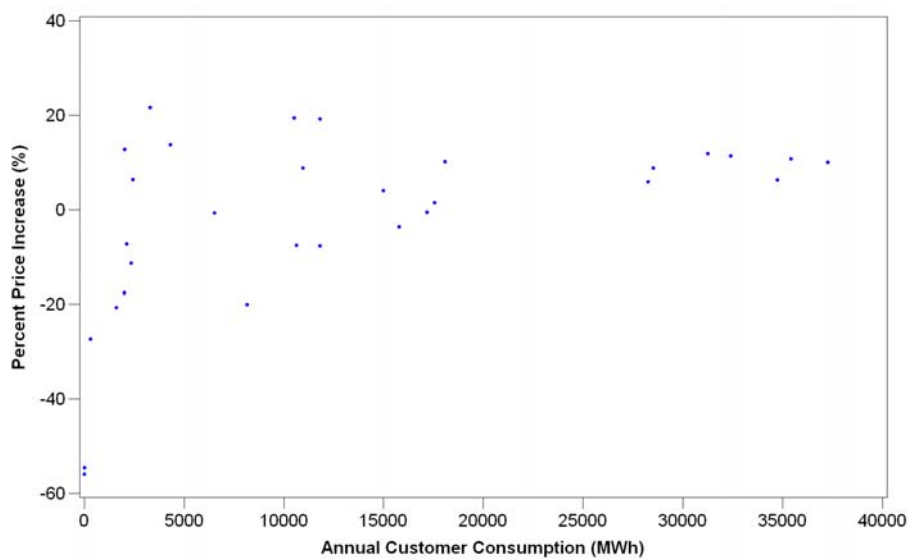
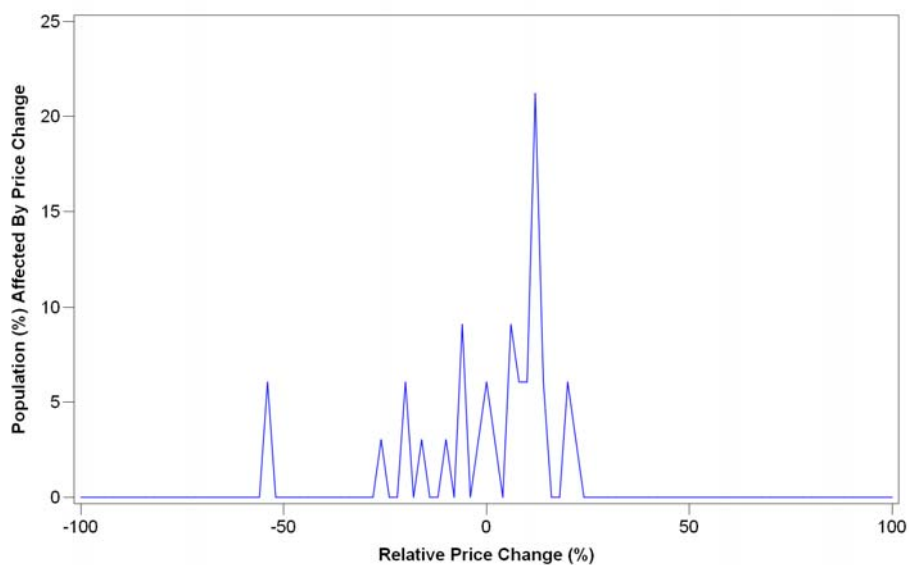


Figure 20 Price change histogram for a sample of EA390 customers



8. Changes from the Previous Regulatory Year (continued)

Table 10 sets out the pricing proposal requirements and where in this pricing proposal the requirement is satisfied.

unders balances have been included in the compliance spreadsheet (Attachment D).

Please note that the quantities for the D-factor scheme, the reasonable estimates scheme, and the transmission overs and

Table 13 Pricing proposal compliance with the rules

Rule requirement	Description	Reference
6.18.2(b)(1)	Set out tariff classes	Chapter 2
6.18.2(b)(2)	Set out proposed tariffs for each tariff classes	Chapter 3
6.18.2(b)(3)	Set out charging parameters and the elements of service	Chapter 3
6.18.2(b)(4)	Set out the weighted average revenue for each tariff class	Chapter 4
6.18.2(b)(5)	Set out the nature of any variation or adjustment to tariffs that could occur during the regulatory year	Chapter 5
6.18.2(b)(5A)	Set out the amount of the Climate Change Fund	Chapter 6
6.18.2(b)(6)	Set out how TUOS charges are passed through	
6.18.2(b)(7)	Compliance of the pricing proposal with the Rules	Chapter 8
6.18.2(b)(8)	Describe changes from the previous regulatory year and compliance of changes to the Rules and the Determination	Chapter 8
6.18.3	Requirements relating to the definition of tariff classes	Chapter 2
6.18.5(a)	Expected revenue must lie between standalone and avoidable costs	Chapter 4
6.18.5(b)	Must take into account LRMC, transactions costs and the ability of customers to respond to price signals when setting tariffs and charging parameters	Chapter 3
6.18.5(c)	Minimum distortion to efficient use to ensure recovery of expected revenue	Chapter 3
6.18.6	Side constraint limitations	Chapter 8
6.18.7	Provision for TUOS charges to be passed onto customers	Chapter 7
6.18.9(a)(3)	Network price trends over the regulatory period	Chapter 8

Attachments

Attachments have been included with this Pricing Proposal as follows:

- Attachment A (CONFIDENTIAL) contains price and tariff class information for Cost Reflective Network Price (CRNP);
- Attachment B (CONFIDENTIAL) contains customer bill information for public lighting customers;
- Attachment C contains price and tariff class information for public lighting customers;
- Attachment D contains the completed compliance spreadsheet;
- Attachment E contains price and tariff class information for Miscellaneous and Monopoly charges;
- Attachment F contains the confirmation of the amount of the Climate Change Fund;
- Attachment G contains details of EnergyAustralia's policy for the assignment and re-assignment of customers to tariffs;
- Attachment H contains a statement from NERA Economic Consulting;
- Attachment I (CONFIDENTIAL) contains a statement from PricewaterhouseCoopers Australia;
- Attachment J contains TransGrid's confirmation of its revenue recovery for the 2009/10 financial year;
- Attachment K contains the Network Tariff Model User Guide;
- Attachment L (CONFIDENTIAL) Economic Cost Model

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